

4

The Hartree–Fock method

4.1 Introduction

Here and in the following chapter we treat two different approaches to the many-electron problem: the Hartree–Fock theory and the density functional theory. Both theories are simplifications of the full problem of many electrons moving in a potential field. In fact, the physical systems we want to study, such as atoms, molecules and solids, consist not only of electrons but also of nuclei, and each of these particles moves in the field generated by the others. A first approximation is to consider the nuclei as being fixed, and to solve the Schrödinger equation for the electronic system in the field of the static nuclei. This approach, called the Born–Oppenheimer approximation, is justified by the nuclei being much heavier than the electrons so that they move at much slower speeds. It remains then to solve for the electronic structure.

The Hartree–Fock method can be viewed as a variational method in which the wave functions of the many-electron system have the form of an antisymmetrised product of one-electron wave functions (the antisymmetrisation is necessary because of the fermion character of the electrons). This restriction leads to an effective Schrödinger equation for the individual one-electron wave functions (called *orbitals*) with a potential determined by the orbitals occupied by the other electrons. This coupling between the orbitals via the potentials causes the resulting equations to become nonlinear in the orbitals, and the solution must be found iteratively in a self-consistency procedure. The Hartree–Fock (HF) procedure is close in spirit to the mean-field approach in statistical mechanics.

We shall see that in this variational approach, correlations between the electrons are neglected to some extent. In particular, the Coulomb repulsion between the electrons is represented in an averaged way. However, the effective interaction caused by the fact that the electrons are fermions, obeying Pauli’s principle, and hence avoid each other if they have the same spin, is accurately included in the HF

approach. There exist several methods that improve on the approximations made in the HF method.

The Hartree–Fock approach is very popular among chemists, and it has also been applied to solids. In this chapter, we give an introduction to the Hartree–Fock method and apply it to simple two-electron systems: the helium atom and the hydrogen molecule. We describe the Born–Oppenheimer approach and independent particle approaches (of which HF is an example) in a bit more detail in the next section. In [Section 4.3](#) we then derive the Hartree method for a two-electron system (the helium atom). In [Section 4.3.2](#), a program for calculating the ground state of the helium atom is described.

In [Sections 4.4](#) and [4.5](#) the HF method for systems containing more than two electrons is described in detail, and in [Section 4.6](#) the basis functions used for molecular systems are described. In [sections 4.7](#) and [4.8](#) some details concerning the implementation of the HF method are considered. In [Section 4.9](#), results of the HF method are presented, and in [Section 4.10](#) the configuration interaction (CI) method, which improves on the HF method is described.

4.2 The Born–Oppenheimer approximation and the independent-particle method

The Hamiltonian of a system consisting of N electrons and K nuclei with charges Z_n reads

$$H = \sum_{i=1}^N \frac{p_i^2}{2m} + \sum_{n=1}^K \frac{p_n^2}{2M_n} + \frac{1}{4\pi\epsilon_0} \frac{1}{2} \sum_{i,j=1; i \neq j}^N \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - \frac{1}{4\pi\epsilon_0} \sum_{n=1}^K \sum_{i=1}^N \frac{Z_n e^2}{|\mathbf{r}_i - \mathbf{R}_n|} + \frac{1}{4\pi\epsilon_0} \frac{1}{2} \sum_{n,n'=1; n \neq n'}^K \frac{Z_n Z_{n'} e^2}{|\mathbf{R}_n - \mathbf{R}_{n'}|}. \quad (4.1)$$

The index i refers to the electrons and n to the nuclei, m is the electron mass, and M_n are the masses of the different nuclei. The first two terms represent the kinetic energies of the electrons and nuclei respectively; the third term represents the Coulomb repulsion between the electrons and the fourth term the Coulomb attraction between electrons and nuclei. Finally, the last term contains the Coulomb repulsion between the nuclei. The wave function of this system depends on the positions $\mathbf{r}_i$ and $\mathbf{R}_n$ of the electrons and nuclei respectively. This Hamiltonian looks quite complicated, and in fact it turns out that if the number of electrons and nuclei is not extremely small (typically smaller than four), it is impossible to solve the stationary Schrödinger equation for this Hamiltonian directly on even the largest and fastest computer available.

Therefore, important approximations must be made, and a first step consists of separating the degrees of freedom connected with the motion of the nuclei from those of the electrons. This procedure is known as the Born–Oppenheimer approximation [1] and its justification resides in the fact that the nuclei are much heavier than the electrons (the mass of a proton or neutron is about 1835 times as large as the electron mass) so it is intuitively clear that the nuclei move much more slowly than the electrons. The latter will then be able to adapt themselves to the current configuration of nuclei. This approach results also from formal calculations (see Problem 4.9), and leads to a Hamiltonian for the electrons in the field generated by a static configuration of nuclei, and a separate Schrödinger equation for the nuclei in which the electronic energy enters as a potential. The Born–Oppenheimer Hamiltonian for the electrons reads

$$H_{\text{BO}} = \sum_{i=1}^N \frac{p_i^2}{2m} + \frac{1}{2} \frac{1}{4\pi\epsilon_0} \sum_{i,j=1; i \neq j}^N \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} - \frac{1}{4\pi\epsilon_0} \sum_{n=1}^K \sum_{i=1}^N \frac{Z_n e^2}{|\mathbf{r}_i - \mathbf{R}_n|}. \quad (4.2)$$

The total energy is the sum of the energy of the electrons and the energy resulting from the Schrödinger equation satisfied by the nuclei. In a further approximation, the motion of the nuclei is neglected and only the electrostatic energy of the nuclei should be added to the energy of the electrons to arrive at the total energy. The positions of the nuclei can be varied in order to find the minimum of this energy, that is, the ground state of the whole system (within the Born–Oppenheimer approximation with static nuclei). In this procedure, the nuclei are treated on a classical footing since their ground state is determined as the minimum of their potential energy, neglecting quantum fluctuations.¹

Even with the positions of the nuclei kept fixed, the problem of solving for the electronic wave functions using the Hamiltonian (4.2) remains intractable, even on a computer, since too many degrees of freedom are involved. It is the second term containing the interactions between the electrons that makes the problem so difficult. If this term were not present, we would be dealing with a sum of one-electron Hamiltonians which can be solved relatively easily. There exist several ways of approximating the eigenfunctions of the Hamiltonian (4.2). In these approaches, the many-electron problem is reduced to an uncoupled problem in which the interaction of one electron with the remaining ones is incorporated in an averaged way into a potential felt by the electron.

¹ Vibrational modes of the nuclei can, however, be treated after expanding the total energy in deviations of the nuclear degrees of freedom from the ground state configuration. A transformation to normal modes then gives us a system consisting of independent harmonic oscillators.

The resulting uncoupled or *independent-particle* (IP) Hamiltonian has the form

$$H_{\text{IP}} = \sum_{i=1}^N \left[\frac{p_i^2}{2m} + V(\mathbf{r}_i) \right]. \quad (4.3)$$

$V(\mathbf{r})$ is a potential depending on the positions $\mathbf{R}_i$ of the nuclei. As we shall see, its form can be quite complicated; in particular, V depends on the wave function ψ on which the IP Hamiltonian is acting. Moreover, V is often a nonlocal operator which means that the value of $V\psi$, evaluated at position $\mathbf{r}$, is determined by the values of ψ at other positions $\mathbf{r}' \neq \mathbf{r}$, and V depends on the energy in some approaches. These complications are the price we have to pay for an independent electron picture.

In the remaining sections of this chapter we shall study the Hartree–Fock approximation and in the next chapter we shall discuss the density functional theory. We start by considering the the helium atom to illustrate the general techniques which will be developed in later sections.

4.3 The helium atom

4.3.1 Self-consistency

In this section, we find an approximate independent-particle Hamiltonian (4.3) for the helium atom within the Born–Oppenheimer approximation by restricting the electronic wave function to a simple form. The coordinates of the wave function are $\mathbf{x}_1$ and $\mathbf{x}_2$, which are combined position and spin coordinates: $\mathbf{x}_i = (\mathbf{r}_i, s_i)$. As electrons are fermions, the wave function must be antisymmetric in the two coordinates $\mathbf{x}_1$ and $\mathbf{x}_2$ (more details concerning antisymmetry and fermions will be given in Section 4.4). We use the following antisymmetric trial wave function for the ground state:

$$\Psi(\mathbf{r}_1, s_1; \mathbf{r}_2, s_2) = \phi(\mathbf{r}_1)\phi(\mathbf{r}_2) \frac{1}{\sqrt{2}}[\alpha(s_1)\beta(s_2) - \alpha(s_2)\beta(s_1)], \quad (4.4)$$

where $\alpha(s)$ denotes the spin-up and $\beta(s)$ the spin-down wave function and ϕ is an orbital – a function depending on a single spatial coordinate – which is shared by the two electrons.

The Born–Oppenheimer Hamiltonian (4.2) for the helium atom reads

$$H_{\text{BO}} = -\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 + \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} - \frac{2}{r_1} - \frac{2}{r_2}, \quad (4.5)$$

where we have used atomic units introduced in Section 3.2.2. We now let this Hamiltonian act on the wave function (4.4). Since the Hamiltonian does not act on the spin, the spin-dependent part drops out on the left and right hand side of the

Schrödinger equation and we are left with:²

$$\left[-\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 - \frac{2}{r_1} - \frac{2}{r_2} + \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \right] \phi(\mathbf{r}_1)\phi(\mathbf{r}_2) = E\phi(\mathbf{r}_1)\phi(\mathbf{r}_2). \quad (4.6)$$

In order to arrive at a simpler equation we remove the $\mathbf{r}_2$ -dependence by multiplying both sides from the left by $\phi^*(\mathbf{r}_2)$ and by integrating over $\mathbf{r}_2$. We then arrive at

$$\left[-\frac{1}{2}\nabla_1^2 - \frac{2}{r_1} + \int d^3r_2 |\phi(\mathbf{r}_2)|^2 \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \right] \phi(\mathbf{r}_1) = E'\phi(\mathbf{r}_1), \quad (4.7)$$

where several integrals yielding a constant (i.e. not dependent on $\mathbf{r}_1$) are absorbed in E' . The third term on the left hand side is recognised as the Coulomb energy of particle 1 in the electric field generated by the charge density of particle 2. To obtain this equation we have used the fact that ϕ is normalised to unity and this normalisation is from now on implicitly assumed for ϕ as occurring in the integral on the left hand side of (4.7). The effective Hamiltonian acting on the orbital of particle 1 has the independent particle form of Eq. (4.3). A remarkable feature is the dependence of the potential on the wave function we are searching for.

Equation (4.7) has the form of a *self-consistency* problem: ϕ is the solution to the Schrödinger equation but the latter is determined by ϕ itself. To solve an equation of this type, one starts with some trial ground state solution $\phi^{(0)}$ which is used in constructing the potential. Solving the Schrödinger equation with this potential, we obtain a new ground state $\phi^{(1)}$ which is used in turn to build a new potential. This procedure is repeated until the ground state $\phi^{(i)}$ and the corresponding energy $E^{(i)}$ of the Schrödinger equation at step i do not deviate appreciably from those in the previous step (if convergence does not occur, we must use some tricks to be discussed in Section 4.7).

The wave function we have used is called *uncorrelated* because of the fact that the probability $P(\mathbf{r}_1, \mathbf{r}_2)$ for finding an electron at $\mathbf{r}_1$ and another one at $\mathbf{r}_2$ is uncorrelated, i.e. it can be written as a product of two one-electron probabilities:

$$P(\mathbf{r}_1, \mathbf{r}_2) = p(\mathbf{r}_1)p(\mathbf{r}_2). \quad (4.8)$$

This does not mean that the electrons do not feel each other: in the determination of the spatial function ϕ , the interaction term $1/|\mathbf{r}_1 - \mathbf{r}_2|$ has been taken into account. But this interaction has been taken into account in an averaged way: it is not the *actual* position of $\mathbf{r}_2$ that determines the wave function for electron 1, but the *average* charge distribution of electron 2. This approach bears much relation to the mean field theory approach in statistical mechanics.

² This equation cannot be satisfied exactly with the form of trial function chosen, as the left hand side depends on $\mathbf{r}_1 - \mathbf{r}_2$ whereas the right hand side does not. We are, however, after the optimal wave function within the set of functions of the form (4.4) in a variational sense, along the lines of the previous chapter, but we want to avoid the complications of carrying out the variational procedure formally. This will be done in Section 4.5.2 for arbitrary numbers of electrons.

The neglect of correlations sometimes leads to unphysical results. An example is found in the dissociation of the hydrogen molecule. Suppose the nuclei are placed at positions $\mathbf{R}_A$ and $\mathbf{R}_B$ and we approximate the one-electron orbitals by spherically symmetric (1s) basis orbitals centred on the two nuclei: $u(\mathbf{r} - \mathbf{R}_A)$ and $u(\mathbf{r} - \mathbf{R}_B)$. Because of the symmetry of the hydrogen molecule, the ground state orbital solution of the independent particle Hamiltonian is given by the symmetric combination of these two basis orbitals:

$$\phi(\mathbf{r}) = u(\mathbf{r} - \mathbf{R}_A) + u(\mathbf{r} - \mathbf{R}_B). \quad (4.9)$$

The total wave function, which contains the product $\phi(\mathbf{r}_1)\phi(\mathbf{r}_2)$ therefore contains ionic terms in which both electrons sit on the same nucleus. This is not so disastrous if the two nuclei are close, but if we separate them, these terms should not be present: they contain the wrong physics and they result in a serious over-estimation of the energy. Physically, this is caused by the fact that electron 1 in our present approximation feels the potential resulting from the *average* charge distribution of electron 2, which is symmetrically distributed over the two nuclei, and thus it ends up on *A* or *B* with equal probability. If electron 1 was to feel the *actual* potential caused by electron 2, it would end up on a different nucleus from electron 2. A better description of the state would therefore be

$$\psi(\mathbf{r}_1; \mathbf{r}_2) = \frac{1}{2}[u(\mathbf{r}_1 - \mathbf{R}_A)u(\mathbf{r}_2 - \mathbf{R}_B) + u(\mathbf{r}_2 - \mathbf{R}_A)u(\mathbf{r}_1 - \mathbf{R}_B)], \quad (4.10)$$

which must then be multiplied by an antisymmetric spin wave function. This wave function is however not of the form (4.4).

The fact that the spatial part of the wave function is equal for the two electrons is specific for the case of two electrons: the antisymmetry is taken care of by the spin part of the wave function. If there are more than two electrons, the situation becomes more complicated and requires much more bookkeeping; this case will be treated in [Section 4.5](#). Neglecting the antisymmetry requirement, one can, however, generalise the results obtained for two electrons to systems with more electrons. Writing the wave function as a product of spin-orbitals $\psi_k(\mathbf{x})$ (spin-orbitals are functions depending on the spatial and spin coordinates of one electron), the following equation for these spin-orbitals is obtained:

$$\left[-\frac{1}{2}\nabla^2 - \sum_n \frac{Z_n}{|\mathbf{r} - \mathbf{R}_n|} + \sum_{l=1}^N \int d\mathbf{x}' |\psi_l(\mathbf{x}')|^2 \frac{1}{|\mathbf{r} - \mathbf{r}'|} \right] \psi_k(\mathbf{x}) = E' \psi_k(\mathbf{x}). \quad (4.11)$$

Here k and l label the spin-orbitals; $\int d\mathbf{x}'$ denotes a sum over the spin s' and an integral over the spatial coordinate $\mathbf{r}'$: $\int d\mathbf{x}' = \sum_{s'} \int d^3r'$. As the Hamiltonian does not act on the spin-dependent part of the spin-orbitals, ψ_k can be written as a product of a spatial orbital with a one-electron spin wave function. In the last term

on the left hand side we recognise the potential resulting from a charge distribution caused by all the electrons; it is called the *Hartree potential*. There is something unphysical about this term: it contains a coupling between orbital k and itself, since this orbital is included in the electron density, even though an electron clearly does not interact with itself. This can be remedied by excluding k from the sum over l in the Hartree term, but then every orbital feels a different potential. In the next subsection, we shall see that this problem is automatically solved in the Hartree–Fock theory which takes the antisymmetry of the many-electron wave function fully into account. Note that in our discussion of the helium case, we have already taken the self-interaction into account because the electron–electron interaction is half the size of that in (4.11) (after summation over the spin in this equation). Equation (4.11) was derived in 1927 by Hartree [2]; it neglects exchange as well as other correlations.

Before studying the problem of more electrons with an antisymmetric wave function, we shall now describe the construction of a program for actually calculating the solution of Eq. (4.7).

4.3.2 A program for calculating the helium ground state

In this section we construct a program for calculating the ground state energy and wave function for the helium atom. In the previous section we have restricted the form of the wave function to be uncorrelated; here we restrict it even further by writing it as a linear combination of four fixed, real basis functions in the same way as in Section 3.2.2. Let us first consider the form assumed by the Schrödinger equation for the independent particle formulation, Eq. (4.7). The parametrisation

$$\phi(\mathbf{r}) = \sum_{p=1}^4 C_p \chi_p(\mathbf{r}) \quad (4.12)$$

leads directly to

$$\begin{aligned} & \left[-\frac{1}{2} \nabla_1^2 - \frac{2}{r_1} + \sum_{r,s=1}^4 C_r C_s \int d^3 r_2 \chi_r(\mathbf{r}_2) \chi_s(\mathbf{r}_2) \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \right] \sum_{q=1}^4 C_q \chi_q(\mathbf{r}_1) \\ &= E' \sum_{q=1}^4 C_q \chi_q(\mathbf{r}_1). \end{aligned} \quad (4.13)$$

Note that the C_p are real as the functions $\chi_p(\mathbf{r})$ are real. From now on we implicitly assume sums over indices $p, q, \dots$ to run from 1 to the number of basis functions N , which is 4 in our case. Multiplying Eq. (4.13) from the left by $\chi_p(\mathbf{r}_1)$ and integrating

over $\mathbf{r}_1$ leads to

$$\sum_{pq} \left(h_{pq} + \sum_{rs} C_r C_s Q_{prqs} \right) C_q = E' \sum_{pq} S_{pq} C_q \quad (4.14)$$

with

$$h_{pq} = \left\langle \chi_p \left| -\frac{1}{2} \nabla^2 - \frac{2}{r} \right| \chi_q \right\rangle; \quad (4.15a)$$

$$Q_{prqs} = \int d^3 r_1 d^3 r_2 \chi_p(\mathbf{r}_1) \chi_r(\mathbf{r}_2) \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \chi_q(\mathbf{r}_1) \chi_s(\mathbf{r}_2) \quad \text{and} \quad (4.15b)$$

$$S_{pq} = \langle \chi_p | \chi_q \rangle. \quad (4.15c)$$

Unfortunately, (4.14) is not a generalised eigenvalue equation because of the presence of the variables C_r and C_s between the brackets on the left hand side. However, if we carry out the self-consistency iteration process as indicated in the previous section, the C_r and C_s are kept *fixed*, and finding the C_q in (4.14) reduces to solving a generalised eigenvalue equation. We then replace C_r , C_s by the solution found and start the same procedure again.

The matrix elements from (4.15) remain to be found. We shall use Gaussian $l = 0$ basis functions (s-functions), just as in the case of the hydrogen atom (see Section 3.2.2). Of course, the optimal exponents α_p occurring in the Gaussian s-basis functions χ_p ,

$$\chi_p(\mathbf{r}) = e^{-\alpha_p r^2}, \quad (4.16)$$

are different from those of the hydrogen atom. Again, rather than solve the non-linear variational problem, which involves not only the prefactors C_p but also the exponents α_p as parameters of the wave function, we shall take the optimal values calculated from a different program which we do not go into here. They are

$$\alpha_1 = 0.298\,073$$

$$\alpha_2 = 1.242\,567$$

$$\alpha_3 = 5.782\,948,$$

$$\alpha_4 = 38.474\,970.$$

The matrix elements of the kinetic and the Coulomb energy are similar to those calculated for the hydrogen atom (see Eq. (3.29)), except for an extra factor of 2 in the nuclear attraction (due to the nuclear charge). In Section 4.8, the matrix element Q_{prqs} will be calculated; the result is given by

$$Q_{prqs} = \frac{2\pi^{5/2}}{(\alpha_p + \alpha_q)(\alpha_r + \alpha_s)\sqrt{\alpha_p + \alpha_q + \alpha_r + \alpha_s}}. \quad (4.17)$$

The program is constructed as follows.

- First, the 4×4 matrices h_{pq} , S_{pq} and the $4 \times 4 \times 4 \times 4$ array Q_{pqrs} are calculated.
- Then initial values for C_p are chosen; they can, for example, all be taken to be equal (of course, you are free to choose other initial values – for this simple system most initial values will converge to the correct answer).
- These C -values are used for constructing the matrix F_{pq} given by

$$F_{pq} = h_{pq} + \sum_{rs} Q_{pqrs} C_r C_s. \quad (4.18)$$

It should be kept in mind that the vector $\mathbf{C}$ should always be normalised to unity via the overlap matrix *before* inserting it into Eq. (4.18):

$$\sum_{p,q=1}^4 C_p S_{pq} C_q = 1 \quad (4.19)$$

(see Problem 3.2).

- Now the generalised eigenvalue problem

$$\mathbf{FC} = E' \mathbf{SC} \quad (4.20)$$

is solved. For the ground state, the vector $\mathbf{C}$ is the one corresponding to the *lowest* eigenvalue.

- The energy for the state found is not simply given by E' as follows from the derivation of the self-consistent Schrödinger equation, Eq. (4.7). The ground state energy can be found by evaluating the expectation value of the Hamiltonian for the ground state just obtained:

$$E_G = 2 \sum_{pq} C_p C_q h_{pq} + \sum_{pqrs} Q_{pqrs} C_p C_q C_r C_s, \quad (4.21)$$

where the (normalised) eigenvector $\mathbf{C}$ results from the last diagonalisation of $\mathbf{F}$.

- The solution $\mathbf{C}$ of the generalised eigenvalue problem (4.20) is then used to build the matrix $\mathbf{F}$ again and so on.

PROGRAMMING EXERCISE

Write a program for calculating the ground state wave function of the helium atom.

Check 1 If your program is correct, the resulting ground state energy should be equal to $-2.855\,160\,38$ a.u. (remember that the atomic energy unit is the Hartree, see Section 3.2.2). The effect of using a small basis set can be judged by comparing with the value -2.8616 a.u. resulting from calculations using continuum integration techniques within the framework of the present calculation as described in Chapter 5. The effect of neglecting correlations in our approach results in the deviation from the exact value -2.903 a.u. (very accurate calculations can be performed for systems containing small numbers of electrons [3]).

4.4 Many-electron systems and the Slater determinant

In the helium problem, we could make use of the fact that in the ground state the required antisymmetry is taken care of by the spin part of the wave function, which drops out of the Schrödinger equation. If it is not the ground state we are after, or if more than two electrons are involved, antisymmetry requirements affect the orbital part of the wave function, and in the next two sections we shall consider a more general approach to an independent electron Hamiltonian, taking this antisymmetry into account. In the present section we consider a particular class of antisymmetric many-electron wave functions and in the next section we shall derive the equations obeyed by them.

When considering a many-electron problem, it must be remembered that electrons are identical particles. This is reflected in the form of the Hamiltonian: for example in (4.2), interchanging electrons i and j does not change the Hamiltonian and the same holds for the independent particle Hamiltonian (4.3). We say that the Hamiltonian commutes with the particle-exchange operator, P_{ij} . This operator acts on a many-electron state and it has the effect of interchanging the coordinates of particles i and j . For an N -particle state³ Ψ

$$P_{ij}\Psi(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \dots, \mathbf{x}_j, \dots, \mathbf{x}_N) = \Psi(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_j, \dots, \mathbf{x}_i, \dots, \mathbf{x}_N). \quad (4.22)$$

In this equation, $\mathbf{x}_i$ is again the combined spin and orbital coordinate:

$$\mathbf{x}_i = (\mathbf{r}_i, s_i). \quad (4.23)$$

As P_{ij} is an Hermitian operator which commutes with the Hamiltonian, the eigenstates of the Hamiltonian are simultaneous eigenstates of P_{ij} with real eigenvalues. Furthermore, as $P_{ij}^2 = 1$ (interchanging a pair twice in a state brings the state back to its original form), its eigenvalue is either $+1$ or -1 . It is an experimental fact that for particles with half-integer spin (fermions) the eigenvalue of the permutation operator is always -1 , and for particles with integer spin (bosons) it is always $+1$. In the first case, the wave function is antisymmetric with respect to particle exchange and in the second case it is symmetric with respect to this operation. As electrons have spin-1/2, the wave function of a many-electron system is antisymmetric with respect to particle exchange.

Let us forget about antisymmetry for a moment. For the case of an independent-particle Hamiltonian, which is a sum of one-electron Hamiltonians as in (4.3), we can write the solution of the Schrödinger equation as a product of one-electron

³ In order to clarify the role of the coordinate s_i , we note that for a single electron the wave function can be written as a *two-spinor*, that is, a two-dimensional vector, and s , which is a two-valued coordinate, selects one component of this spinor. When dealing with more particles (N), the two-spinors combine into a 2^N -dimensional one and the combined coordinates $s_1, \dots, s_N$ select a component of this large spinor (which depends on the positions $\mathbf{r}_i$).

states:

$$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N) = \psi_1(\mathbf{x}_1) \cdots \psi_N(\mathbf{x}_N). \quad (4.24)$$

The one-electron states ψ_k are eigenstates of the one-particle Hamiltonian, so they are orthogonal. The probability density for finding the particles with specific values $\mathbf{x}_1, \dots, \mathbf{x}_N$ of their coordinates is given by

$$\rho(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N) = |\psi_1(\mathbf{x}_1)|^2 |\psi_2(\mathbf{x}_2)|^2 \cdots |\psi_N(\mathbf{x}_N)|^2, \quad (4.25)$$

which is just the product of the one-electron probability densities. Such a probability distribution is called *uncorrelated*, and therefore we will use the term ‘uncorrelated’ for the wave function in (4.24) too.

Of course, the same state as (4.24) but with the spin-orbitals permuted, is a solution too, as are linear combinations of several such states. But we require antisymmetric states, and an antisymmetric linear combination of a minimal number of terms of the form (4.24) is given by

$$\Psi_{AS}(\mathbf{x}_1, \dots, \mathbf{x}_N) = \frac{1}{\sqrt{N!}} \sum_{\mathbf{P}} \epsilon_P \mathbf{P} \psi_1(\mathbf{x}_1) \cdots \psi_N(\mathbf{x}_N). \quad (4.26)$$

$\mathbf{P}$ is a permutation operator which permutes the *coordinates* of the spin-orbitals only, and not their labels (if $\mathbf{P}$ acted on the latter too, it would have no effect at all!); alternatively, one could have $\mathbf{P}$ acting on the labels only, the choice is merely a matter of convention. The above-mentioned exchange operator is an example of this type of operator. In (4.26), all permutations are summed over and the states are multiplied by the sign ϵ_P of the permutation (the sign is +1 or -1 according to whether the permutation can be written as product of an even or an odd number of pair interchanges respectively).

We can write (4.26) in the form of a *Slater determinant*:

$$\Psi_{AS}(\mathbf{x}_1, \dots, \mathbf{x}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\mathbf{x}_1) & \psi_2(\mathbf{x}_1) & \cdots & \psi_N(\mathbf{x}_1) \\ \psi_1(\mathbf{x}_2) & \psi_2(\mathbf{x}_2) & \cdots & \psi_N(\mathbf{x}_2) \\ \vdots & \vdots & & \vdots \\ \psi_1(\mathbf{x}_N) & \psi_2(\mathbf{x}_N) & \cdots & \psi_N(\mathbf{x}_N) \end{vmatrix}. \quad (4.27)$$

It is important to note that after this antisymmetrisation procedure the electrons are correlated. To see this, consider the probability density of finding one electron with coordinates $\mathbf{x}_1$ and another with $\mathbf{x}_2$:

$$\begin{aligned} \rho(\mathbf{x}_1, \mathbf{x}_2) &= \int d\mathbf{x}_3 \cdots d\mathbf{x}_N |\Psi_{AS}(\mathbf{x}_1, \dots, \mathbf{x}_N)|^2 \\ &= \frac{1}{N(N-1)} \sum_{k,l} [|\psi_k(\mathbf{x}_1)|^2 |\psi_l(\mathbf{x}_2)|^2 - \psi_k^*(\mathbf{x}_1) \psi_k(\mathbf{x}_2) \psi_l^*(\mathbf{x}_2) \psi_l(\mathbf{x}_1)]. \end{aligned} \quad (4.28)$$

To find the probability of finding two electrons at positions $\mathbf{r}_1$ and $\mathbf{r}_2$, we must sum over the spin variables:

$$\rho(\mathbf{r}_1, \mathbf{r}_2) = \sum_{s_1, s_2} \rho(\mathbf{x}_1, \mathbf{x}_2). \quad (4.29)$$

For spin-orbitals that can be written as a product of a spatial orbital and a one-particle spin wave function, it is seen that for ψ_k and ψ_l having opposite spin, the second term vanishes and therefore opposite spin-orbitals are still uncorrelated (the first term of (4.28) obviously describes uncorrelated probabilities) but for equal spins, the two terms cancel when $\mathbf{r}_1 = \mathbf{r}_2$, so we see that electron pairs with parallel spin are kept apart. Every electron is surrounded by an ‘exchange hole’ [4] in which other electrons having the same spin are hardly found. Comparing (4.28) with the uncorrelated form (4.25), we see that exchange introduces correlation effects. However, the term ‘correlation effects’ is usually reserved for all correlations *apart from* exchange.

It is possible to construct Slater determinants from general spin-orbitals, i.e. not necessarily eigenstates of the one-electron Hamiltonian. It is even possible to take these spin-orbitals to be nonorthogonal. However, if there is overlap between two such spin-orbitals, this drops out in constructing the Slater determinant. Therefore we shall take the spin-orbitals from which the Slater determinant is constructed to be orthonormal.

Single Slater determinants form a basis in the space of all antisymmetric wave functions. In Section 4.10, we shall describe a method in which this fact is used to take correlations into account.

4.5 Self-consistency and exchange: Hartree–Fock theory

4.5.1 The Hartree–Fock equations – physical picture

Fock extended the Hartree equation (4.11) by taking antisymmetry into account. We first give the result which is known as the Hartree–Fock equation; the full derivation is given in Section 4.5.2 [5, 6]:

$$\mathcal{F}\psi_k = \epsilon_k \psi_k \text{ with} \quad (4.30)$$

$$\begin{aligned} \mathcal{F}\psi_k = & \left[-\frac{1}{2}\nabla^2 - \sum_n \frac{Z_n}{|\mathbf{r} - \mathbf{R}_n|} \right] \psi_k(\mathbf{x}) + \sum_{l=1}^N \int d\mathbf{x}' |\psi_l(\mathbf{x}')|^2 \frac{1}{|\mathbf{r} - \mathbf{r}'|} \psi_k(\mathbf{x}) \\ & - \sum_{l=1}^N \int d\mathbf{x}' \psi_l^*(\mathbf{x}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} \psi_k(\mathbf{x}') \psi_l(\mathbf{x}). \end{aligned} \quad (4.31)$$

The operator $\mathcal{F}$ is called the *Fock operator*. The first three terms on the right hand side are the same as in those appearing in the Hartree equation. The fourth term is

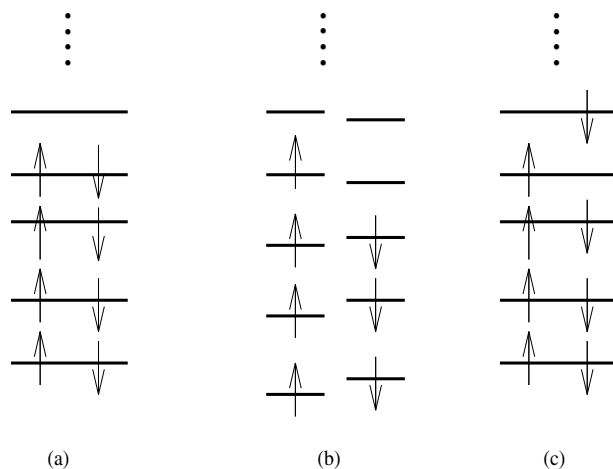


Figure 4.1. The Hartree–Fock spectrum. The figure shows how the levels are filled for (a) the ground state of an even number of electrons, (b) the ground state of an odd number of electrons and (c) an excited state in the spectrum of (a). Note that the spectrum in (c) does not correspond to the ground state; see [Section 4.5.3](#). Instead it corresponds to the restricted approximation, in which the same set of energy levels is available for electrons with both spins.

the same as the third, with two spin-orbital labels k and l interchanged and a minus sign in front resulting from the antisymmetry of the wave function – it is called the exchange term. Note that this term is nonlocal: it is an operator acting on ψ_k , but its value at $\mathbf{r}$ is determined by the value assumed by ψ_k at all possible positions $\mathbf{r}'$.

A subtlety is that the eigenvalues ϵ_k of the Fock operator are not the energies of single electron orbitals, although they are related to the total energy by

$$E = \frac{1}{2} \sum_k [\epsilon_k + \langle \psi_k | h | \psi_k \rangle]. \quad (4.32)$$

In [Section 4.5.3](#) we shall see that the individual levels ϵ_k can be related to excitation energies within some approximation.

It is clear that (4.31) is a nonlinear equation, which must be solved by a self-consistency iterative procedure analogously to the previous section. Sometimes the name ‘self-consistent field theory’ (SCF) is used for this type of approach. The self-consistency procedure is carried out as follows. Solving (4.31) yields an infinite spectrum. To find the ground state, we must take the lowest N eigenstates of this spectrum as the spin-orbitals of the electrons. These are the ψ_l which are then used to build the new Fock operator which is diagonalised again and the procedure is repeated over and over until convergence is achieved. [Figure 4.1\(a\)](#) and [\(b\)](#) gives a schematic representation of the Hartree–Fock spectrum and shows how the levels

are filled. Of course, it is not clear a priori that the lowest energy of the system is found by filling the lowest states of the Fock spectrum because the energy is not simply a sum over the Fock eigenvalues. However, in practical applications this turns out to be the case.

The Hartree–Fock theory is the cornerstone of electronic structure calculations for atoms and molecules. There exists a method, configuration interaction, which provides a systematic way of improving upon Hartree–Fock theory; it will be described briefly in [Section 4.10](#). In solid state physics, density functional theory is used mostly instead of Hartree–Fock theory (see [Chapter 5](#)).

The exchange term in (4.31) is a direct consequence of the particle exchange-antisymmetry of the wave function. It vanishes for orthogonal states k and l , so pairs with opposite spin do not feel this term. The self-energy problem with the Hartree potential mentioned at the end of [Section 4.3.1](#) appears to be solved in the Hartree–Fock equations: the self-energy term in the Hartree energy is cancelled by the exchange contribution as a result of the antisymmetry.

The exchange contribution *lowers* the Coulomb interaction between the electrons, which can be viewed as a consequence of the fact that exchange keeps electrons with the same spin apart; see the discussion below [Eq. \(4.28\)](#). The dependence of this change in Coulomb energy on the electron density can be estimated using a simple classical argument. Suppose that in an electron gas with average density n , each electron occupies a volume which is not accessible to other electrons with like spin. This volume can be approximated by a sphere with radius $r_c \propto n^{-1/3}$. Comparing the Coulomb interaction per volume for such a system with one in which the electrons are distributed homogeneously throughout space, we obtain

$$\Delta E_C \approx n^2 \left[\int_{r_c}^{\infty} r^2 dr \frac{1}{r} - \int_0^{\infty} r^2 dr \frac{1}{r} \right] = -n^2 \int_0^{r_c} r^2 dr \frac{1}{r} \propto -n^2 r_c^2 \propto n^{4/3}. \quad (4.33)$$

One of the two factors n in front of the integral comes from the average density seen by one electron, and the second factor counts how many electrons per volume experience this change in electrostatic energy. The $n^{4/3}$ dependence of the exchange contribution is also found in more sophisticated derivations [\[7\]](#) and we shall meet it again when discussing the local density approximation in the next chapter.

*4.5.2 Derivation of the Hartree–Fock equations

The derivation of the Fock equation consists of performing a variational calculation for the Schrödinger equation, where the subspace to which we shall confine ourselves is the space of all single Slater determinants like [Eq. \(4.27\)](#). We must

therefore calculate the expectation value of the energy for an arbitrary Slater determinant using the Born–Oppenheimer Hamiltonian and then minimise the result with respect to the spin-orbitals in the determinant.

We write the Hamiltonian as follows:

$$\begin{aligned}
 H &= \sum_i h(i) + \frac{1}{2} \sum_{i,j; i \neq j} g(i,j) \quad \text{with} \\
 g(i,j) &= \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} \quad \text{and} \\
 h(i) &= -\frac{1}{2} \nabla_i^2 - \sum_n \frac{Z_n}{|\mathbf{r}_i - \mathbf{R}_n|}.
 \end{aligned} \tag{4.34}$$

$h(i)$ depends on $\mathbf{r}_i$ only and $g(i,j)$ on $\mathbf{r}_i$ and $\mathbf{r}_j$. Writing the Slater determinant ψ as a sum of products of spin-orbitals and using the orthonormality of the latter, it can easily be verified that this determinant is normalised, and for the matrix element of the one-electron part of the Hamiltonian, we find (see Problem 4.3)

$$\begin{aligned}
 \left\langle \Psi_{\text{AS}} \left| \sum_i h(i) \right| \Psi_{\text{AS}} \right\rangle &= N \cdot \frac{(N-1)!}{N!} \sum_k \langle \psi_k | h | \psi_k \rangle \\
 &= \sum_k \langle \psi_k | h | \psi_k \rangle = \sum_k \int d\mathbf{x} \, \psi_k^*(\mathbf{x}) h(\mathbf{r}) \psi_k(\mathbf{x}).
 \end{aligned} \tag{4.35}$$

By $\int d\mathbf{x}$ we denote an integral over the spatial coordinates and a sum over the spin-degrees of freedom as usual.

The matrix element of the two-electron term $g(i,j)$ for a Slater determinant not only gives a nonzero contribution when the spin-orbitals in the left and right hand sides of the inner product occur in the same order, but also for k and l interchanged on one side (the derivation is treated in Problem 4.3):

$$\left\langle \Psi_{\text{AS}} \left| \sum_{i,j} g(i,j) \right| \Psi_{\text{AS}} \right\rangle = \sum_{k,l} \langle \psi_k \psi_l | g | \psi_k \psi_l \rangle - \sum_{k,l} \langle \psi_k \psi_l | g | \psi_l \psi_k \rangle. \tag{4.36}$$

In this equation, the following notation is used:

$$\langle \psi_k \psi_l | g | \psi_m \psi_n \rangle = \int d\mathbf{x}_1 d\mathbf{x}_2 \, \psi_k^*(\mathbf{x}_1) \psi_l^*(\mathbf{x}_2) \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \psi_m(\mathbf{x}_1) \psi_n(\mathbf{x}_2). \tag{4.37}$$

In summary, we obtain for the expectation value of the energy:

$$E = \sum_k \langle \psi_k | h | \psi_k \rangle + \frac{1}{2} \sum_{kl} [\langle \psi_k \psi_l | g | \psi_k \psi_l \rangle - \langle \psi_k \psi_l | g | \psi_l \psi_k \rangle]. \tag{4.38}$$

We now define the operators

$$J_k(\mathbf{x})\psi(\mathbf{x}) = \int \psi_k^*(\mathbf{x}') \frac{1}{r_{12}} \psi_k(\mathbf{x}') \psi(\mathbf{x}) d\mathbf{x}' \quad \text{and} \quad (4.39a)$$

$$K_k(\mathbf{x})\psi(\mathbf{x}) = \int \psi_k^*(\mathbf{x}') \frac{1}{r_{12}} \psi(\mathbf{x}') \psi_k(\mathbf{x}) d\mathbf{x}' \quad (4.39b)$$

and furthermore

$$J = \sum_k J_k; \quad K = \sum_k K_k. \quad (4.40)$$

J is called the *Coulomb* operator and K the *exchange* operator as it can be obtained from the Coulomb operator by interchanging the two rightmost spin-orbitals. In terms of these operators, we can write the energy as

$$E = \sum_k \left\langle \psi_k \left| h + \frac{1}{2}(J - K) \right| \psi_k \right\rangle. \quad (4.41)$$

This is the energy-functional for a Slater determinant. We determine the minimum of this functional as a function of the spin-orbitals ψ_k , and the spin-orbitals for which this minimum is assumed give us the many-electron ground state. Notice however that the variation in the spin-orbitals ψ_k is not completely arbitrary, but should respect the orthonormality relation:

$$\langle \psi_k | \psi_l \rangle = \delta_{kl}. \quad (4.42)$$

This implies that we have a minimisation problem with constraints, which can be solved using the Lagrange multiplier theorem. Note that there are only $N(N+1)/2$ independent constraints as $\langle \psi_k | \psi_l \rangle = \langle \psi_l | \psi_k \rangle^*$. Using the Lagrange multipliers Λ_{kl} for the constraints (4.42), we have

$$\delta E - \sum_{kl} \Lambda_{kl} [\langle \delta \psi_k | \psi_l \rangle - \langle \psi_k | \delta \psi_l \rangle] = 0 \quad (4.43)$$

with

$$\begin{aligned} \delta E &= \sum_k \langle \delta \psi_k | h | \psi_k \rangle + \text{complex conj.} \\ &\quad + \frac{1}{2} \sum_{kl} (\langle \delta \psi_k \psi_l | g | \psi_k \psi_l \rangle + \langle \psi_l \delta \psi_k | g | \psi_l \psi_k \rangle \\ &\quad - \langle \delta \psi_k \psi_l | g | \psi_l \psi_k \rangle - \langle \psi_l \delta \psi_k | g | \psi_k \psi_l \rangle) + \text{complex conj.} \\ &= \sum_k \langle \delta \psi_k | h | \psi_k \rangle + \text{complex conj.} \\ &\quad + \sum_{kl} (\langle \delta \psi_k \psi_l | g | \psi_k \psi_l \rangle - \langle \delta \psi_k \psi_l | g | \psi_l \psi_k \rangle) + \text{complex conj.} \end{aligned} \quad (4.44)$$

where in the second step the following symmetry property of the two-electron matrix elements is used:

$$\langle \psi_k \psi_l | g | \psi_m \psi_n \rangle = \langle \psi_l \psi_k | g | \psi_n \psi_m \rangle. \quad (4.45)$$

Note furthermore that because of the symmetry of the constraint equations, we must have $\Lambda_{kl} = \Lambda_{lk}^*$. Eq. (4.44) can be rewritten as

$$\delta E = \sum_k \langle \delta \psi_k | \mathcal{F} | \psi_k \rangle + \langle \psi_k | \mathcal{F} | \delta \psi_k \rangle \quad (4.46)$$

with

$$\mathcal{F} = h + J - K. \quad (4.47)$$

The Hermitian operator $\mathcal{F}$ is the Fock operator, now formulated in terms of the operators J and K . It is important to note that in this equation, J and K occur with the same prefactor as h , in contrast to Eq. (4.41) in which both J and K have a factor $1/2$ compared with h . This extra factor is caused by the presence of two spin-orbitals in the expressions for J and K which yield extra terms in the derivative of the energy. The matrix elements of the Fock operator with respect to the spin-orbitals ψ_k are

$$\langle \psi_k | \mathcal{F} | \psi_l \rangle = h_{kl} + \sum_{k'} [\langle \psi_k \psi_{k'} | g | \psi_l \psi_{k'} \rangle - \langle \psi_k \psi_{k'} | g | \psi_{k'} \psi_l \rangle]. \quad (4.48)$$

We finally arrive at the equation

$$\langle \delta \psi_k | \mathcal{F} | \psi_k \rangle + \langle \psi_k | \mathcal{F} | \delta \psi_k \rangle + \sum_l \Lambda_{kl} (\langle \delta \psi_k | \psi_l \rangle - \langle \psi_l | \delta \psi_k \rangle) = 0 \quad (4.49)$$

and since $\delta \psi$ is small but arbitrary, this, with $\Lambda_{kl} = \Lambda_{lk}^*$, leads to

$$\mathcal{F} \psi_k = \sum_l \Lambda_{kl} \psi_l. \quad (4.50)$$

The Lagrange parameters Λ_{kl} in this equation cannot be chosen freely: they must be such that the solutions ψ_k form an orthonormal set. An obvious solution of the above equation is found by taking the ψ_k as the eigenvectors of the Fock operators with eigenvalues ϵ_k , and $\Lambda_{kl} = \epsilon_k \delta_{kl}$:

$$\mathcal{F} \psi_k = \epsilon_k \psi_k. \quad (4.51)$$

This equation is the same as (4.31), presented at the beginning of the previous subsection. We can find other solutions to the general Fock equation (4.50) by transforming the set of eigenstates $\{\psi_k\}$ according to a unitary transformation, defined by a (unitary) matrix U :

$$\psi'_k = \sum_l U_{kl} \psi_l. \quad (4.52)$$

The resulting states ψ'_k then form an orthonormal set, satisfying (4.50) with

$$\Lambda_{kl} = \sum_{lm} U_{km} \epsilon_m U_{ml}^\dagger. \quad (4.53)$$

In fact, a unitary transformation of the set $\{\psi_k\}$ leaves the Slater determinant unchanged (see Problem 4.7).

Equation (4.51) has the form of an ordinary Schrödinger equation although the eigenvalues ϵ_k are identified as Lagrange multipliers rather than as energies; nevertheless they are often called ‘orbital energies’. From (4.51) and (4.38) it can be seen that the energy is related to the parameters ϵ_k by

$$E = \frac{1}{2} \sum_k [\epsilon_k + \langle \psi_k | h | \psi_k \rangle] = \sum_k \left[\epsilon_k - \frac{1}{2} \langle k | J - K | k \rangle \right]. \quad (4.54)$$

The second form shows how the Coulomb and exchange contribution must be subtracted from the sum of the Fock levels to avoid counting the two-electron integrals twice.

In the previous section we have already seen how the self-consistency procedure for solving the resulting equations is carried out.

4.5.3 Koopman’s theorem

If we were to calculate an excited state, we would have to take the lowest $N - 1$ spin-orbitals from the Fock spectrum and one excited spin-orbital for example (see Fig. 4.1c), and carry out the self-consistency procedure for this configuration. The resulting eigenstates will differ from the corresponding eigenstates in the ground state. If we assume that the states do not vary appreciably when constructing the Slater determinant from excited spin-orbitals instead of the ground state ones, we can predict excitation energies from a ground state calculation. It turns out that – within the approximation that the spin-orbitals are those of the ground state – the difference between the sums of the eigenvalues, ϵ_k , of the ground state and excited state configuration, is equal to the real energy difference; see Problem 4.6. This is known as *Koopman’s theorem*. This is not really a theorem but a way of approximating excitation energies which turns out to work well for many systems. For further reading, see Refs. [5, 6, 8].

4.6 Basis functions

In the derivation leading to (4.51) (or (4.31)), the possible solutions of the Schrödinger equation were restricted to the space of single Slater determinants. To solve the resulting eigenvalue equation, another variational principle in the same

spirit as in the previous chapter and in [Section 4.3.2](#) can be used, that is, expanding the spin-orbitals ψ_k as linear combinations of a finite number of basis states χ_p :

$$\psi_k(\mathbf{x}) = \sum_{p=1}^M C_{pk} \chi_p(\mathbf{x}). \quad (4.55)$$

Then (4.51) assumes a matrix form

$$\mathbf{FC}_k = \epsilon_k \mathbf{SC}_k \quad (4.56)$$

where $\mathbf{S}$ is the overlap matrix for the basis used.

In the next subsection we shall consider how spin and orbital parts are combined in the basis sets and in [Section 4.6.2](#) we shall discuss the form of the orbital basis functions.

4.6.1 Closed- and open-shell systems

In a *closed-shell* system, the levels are occupied by two electrons with opposite spin whereas in an *open-shell* system there are partially filled levels containing only one electron. If the number of electrons is even, the system does not necessarily have to be closed-shell since there may be degenerate levels (apart from spin-degeneracy) each containing one electron – or we might be considering an excited state in which an electron is pushed up to a higher level. If the number of electrons is odd, the system will always be open-shell.

Consider the addition of an electron to a closed-shell system. The new electron will interact differently with the spin-up and -down electrons present in the system, as exchange is felt by parallel spin pairs only. Therefore, if the levels of the system without the extra electron are spin-up and -down degenerate, they will now split into two sublevels with different orbital dependence, the lower sublevel having the same spin as the new electron.⁴ We see that the spin-up and -down degeneracy of the levels of a closed-shell system is lifted in the open-shell case.

It is important to note that even when the number of electrons is even, the unrestricted solution may be different from the restricted one. To see this, consider again the discussion of the hydrogen molecule in [Section 4.3.1](#). A possible description of the state within the unrestricted scheme is

$$\begin{aligned} \psi(\mathbf{x}_1, \mathbf{x}_2) = & \frac{1}{\sqrt{2}} [u(\mathbf{r}_1 - \mathbf{R}_A)\alpha(s_1)u(\mathbf{r}_2 - \mathbf{R}_B)\beta(s_2) \\ & - u(\mathbf{r}_2 - \mathbf{R}_A)\alpha(s_2)u(\mathbf{r}_1 - \mathbf{R}_B)\beta(s_1)]. \end{aligned} \quad (4.57)$$

⁴ An exception to this rule occurs when the Coulomb interaction between the degenerate levels and the new electron vanishes as a result of symmetry.

This describes two electrons located at different nuclei, which is correct for large nuclear separation, but this is not an eigenstate of the total spin operator. When the nuclei are separated, the state crosses over from a restricted to an unrestricted one. The distance at which this happens is the Coulson–Fisher point [9].

In a closed-shell system the $2N$ orbitals can be grouped in pairs with the same orbital dependence but with opposite spin, thus reflecting the spin-degeneracy:

$$\{\psi_{2k-1}(\mathbf{x}), \psi_{2k}(\mathbf{x})\} = \{\phi_k(\mathbf{r})\alpha(s), \phi_k(\mathbf{r})\beta(s)\}, \quad k = 1, \dots, N. \quad (4.58)$$

The $\phi_k(\mathbf{r})$ are the spatial orbitals and $\alpha(s)$, $\beta(s)$ are the up and down spin-states respectively. For an open-shell system, such pairing does not occur for all levels, and to obtain accurate results, it is necessary to allow for a different orbital dependence for each spin-orbital in most cases. Even for an open-shell system it is possible to impose the restriction (4.58) on the spin-orbitals, neglecting the splitting of the latter, but the results will be less accurate in that case. Calculations with the spin-orbitals paired as in (4.58) are called restricted Hartree–Fock (RHF) and those in which all spin-orbitals are allowed to have a different spatial dependence are called unrestricted Hartree–Fock (UHF). UHF eigenstates are usually inconvenient because they are not eigenstates of the total spin-operator, as can easily be verified by combining two different orbitals with a spin-up and -down function respectively. On the other hand, the energy is more accurate.

We shall now rewrite the Hartree–Fock equations for RHF using the special structure of the set of spin-orbitals given in (4.58). As we have seen in the previous section, the general form of the Fock operator is

$$\mathcal{F} = h + J - K \quad (4.59)$$

with

$$\begin{aligned} J(\mathbf{x})\psi(\mathbf{x}) &= \sum_l \int dx' \psi_l^*(\mathbf{x}') \psi_l(\mathbf{x}') \frac{1}{r_{12}} \psi(\mathbf{x}); \\ K(\mathbf{x})\psi(\mathbf{x}) &= \sum_l \int dx' \psi_l^*(\mathbf{x}') \psi(\mathbf{x}') \frac{1}{r_{12}} \psi_l(\mathbf{x}). \end{aligned} \quad (4.60)$$

The sum over l is over all occupied Fock levels. As the Fock operator depends explicitly on the spatial coordinate only (there is an implicit spin-dependence via the spin-orbitals occurring in the Coulomb and exchange operators), it is possible to eliminate the spin degrees of freedom by summing over them and find an operator acting only on the spatial orbitals $\phi(\mathbf{r})$. The uncoupled single-particle Hamiltonian h remains the same since it contains neither explicit nor implicit spin-dependence, and from (4.60) it is seen that the Coulomb and exchange operators, written in terms

of the orbital parts only, read

$$\begin{aligned}\tilde{J}(\mathbf{r})\phi(\mathbf{r}) &= 2 \sum_l \int d^3 r' \phi_l^*(\mathbf{r}') \phi_l(\mathbf{r}') \frac{1}{|\mathbf{r}' - \mathbf{r}|} \phi(\mathbf{r}) \quad \text{and} \\ \tilde{K}(\mathbf{r})\phi(\mathbf{r}) &= \sum_l \int d^3 r' \phi_l^*(\mathbf{r}') \phi(\mathbf{r}') \frac{1}{|\mathbf{r}' - \mathbf{r}|} \phi_l(\mathbf{r}).\end{aligned}\tag{4.61}$$

In contrast with Eq. (4.60), the sums over l run over half the number of electrons because the spin degrees of freedom have been summed over. The Fock operator now becomes

$$\tilde{\mathcal{F}}(\mathbf{r}) = h(\mathbf{r}) + 2\tilde{J}(\mathbf{r}) - \tilde{K}(\mathbf{r}).\tag{4.62}$$

From now on, we shall only use this spatial form of the Fock operator and drop the tilde from the operators in (4.61) and (4.62). The corresponding expression for the energy is found analogously and is given by

$$E_g = 2 \sum_k \langle \phi_k | h | \phi_k \rangle + \sum_k (2 \langle \phi_k | J | \phi_k \rangle - \langle \phi_k | K | \phi_k \rangle).\tag{4.63}$$

It is possible to solve the Fock equation using a finite basis set, in the same spirit as the helium calculation of Section 4.3.2. The spin part of the basis functions is simply $\alpha(s)$ or $\beta(s)$ (spin-up and -down respectively) and the orbital part $\chi_p(\mathbf{r})$ needs to be specified – this will be done in the next section. For a given basis $\chi_p(\mathbf{r})$, we obtain the following matrix equation, which is known as the *Roothaan equation*:

$$\mathbf{F}\mathbf{C}_k = \epsilon \mathbf{S}\mathbf{C}_k,\tag{4.64}$$

similar to (4.51), but now $\mathbf{S}$ is the overlap matrix for the *orbital* basis $\chi_p(\mathbf{r})$ and the matrix $\mathbf{F}$ is given by

$$F_{pq} = h_{pq} + \sum_k \sum_{rs} C_{rk}^* C_{sk} (2 \langle pr | g | qs \rangle - \langle pr | g | sq \rangle)\tag{4.65}$$

where

$$h_{pq} = \langle p | h | q \rangle = \int d^3 r \chi_p^*(\mathbf{r}) \left[-\frac{1}{2} \nabla^2 - \sum_n \frac{Z_n}{|\mathbf{R}_n - \mathbf{r}|} \right] \chi_q(\mathbf{r}),\tag{4.66}$$

and

$$\langle pr | g | qs \rangle = \int d^3 r_1 d^3 r_2 \chi_p^*(\mathbf{r}_1) \chi_r^*(\mathbf{r}_2) \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} \chi_q(\mathbf{r}_1) \chi_s(\mathbf{r}_2).\tag{4.67}$$

k labels the orbitals ϕ_k and p, q, r and s label the basis functions. Generally, sums over labels k and l run over the occupied orbitals, and sums over p, q, r, s run over the functions in the basis set.

It is convenient to introduce the density matrix which (for RHF) is defined as

$$P_{pq} = 2 \sum_k C_{pk} C_{qk}^*. \quad (4.68)$$

This is the matrix representation of the operator

$$\rho = 2 \sum_k |\phi_k\rangle \langle \phi_k| \quad (4.69)$$

which is recognised as the usual definition of the density matrix in quantum theory (the factor 2 is due to the spin). Using (4.68), the Fock matrix can be written as

$$F_{pq} = h_{pq} + \frac{1}{2} \sum_{rs} P_{sr} [2 \langle pr | g | qs \rangle - \langle pr | g | sq \rangle], \quad (4.70)$$

and the energy is given by

$$E = \sum_{pq} P_{pq} h_{pq} + \frac{1}{2} \sum_{pqrs} P_{pq} P_{sr} \left[\langle pr | g | qs \rangle - \frac{1}{2} \langle pr | g | sq \rangle \right]. \quad (4.71)$$

For the UHF case, it is convenient to define an orbital basis $\chi_p(\mathbf{r})$ and the spin-up orbitals are now represented by the vector $\mathbf{C}^+$ and the spin-down ones by $\mathbf{C}^-$. Using these vectors to reformulate the Hartree–Fock equations, the so-called Pople–Nesbet equations are obtained:

$$\begin{aligned} \mathbf{F}^+ \mathbf{C}^+ &= \epsilon^+ \mathbf{S} \mathbf{C}^+ \\ \mathbf{F}^- \mathbf{C}^- &= \epsilon^- \mathbf{S} \mathbf{C}^- \end{aligned} \quad (4.72)$$

with

$$\begin{aligned} F_{pq}^{\pm} &= h_{pq} + \sum_{k_{\pm}} \sum_{rs} C_{rk_{\pm}}^{\pm*} C_{sk_{\pm}}^{\pm} [\langle pr | g | qs \rangle - \langle pr | g | sq \rangle] \\ &+ \sum_{k_{\mp}} \sum_{rs} C_{rk_{\mp}}^{\mp*} C_{sk_{\mp}}^{\mp} \langle pr | g | qs \rangle. \end{aligned} \quad (4.73)$$

In practice, real orbital basis functions are used, so that complex conjugates can be removed from the C_{pk} in Eqs. (4.65), (4.68) and (4.73). In the following, we shall restrict ourselves to RHF.

4.6.2 Basis functions: STO and GTO

In this subsection, we discuss the basis functions used in the atomic and molecular Hartree–Fock programs. As already noted in Chapter 3, the basis must be chosen carefully: the matrix diagonalisation we must perform scales with the third power of the number of basis functions, so a small basis set is desirable which is able to

model the exact solutions to the Fock equations accurately. A molecule consists of atoms which, in isolation, have a number of atomic orbitals occupied by the electrons. If we put the atoms together in a molecule, the orbitals with low energies will be slightly perturbed by the new environment and the valence electrons will now orbit around more than one nucleus, thus binding the molecule together. In the molecule, the electrons now occupy *molecular orbitals* (MO). In constructing a basis, it turns out to be efficient to start from the atomic orbitals. The one-electron wave functions that can be constructed from these orbitals are linear combinations of atomic orbitals (LCAO). The solutions to the HF equations, which have the form

$$\phi_k(\mathbf{r}) = \sum_p C_{pk} \chi_p(\mathbf{r}), \quad (4.74)$$

are the molecular orbitals, written in LCAO form.

Analytic forms of the atomic orbitals are only known for the hydrogen atom but they can be used for more general systems. The orbitals of the hydrogen atom have the following form:

$$\chi(\mathbf{r}) = f_{n-1}(r) r^{-l} P_l(x, y, z) e^{-r/n}, \quad (4.75)$$

where l is the angular momentum quantum number, P_l is a polynomial in x , y and z of degree l containing the angular dependence; $f_{n-1}(r)$ is a polynomial in r of degree $n - 1$; n is an integer (r is expressed in atomic units a_0). This leads to the following general form of atomic orbital basis functions:

$$\chi_\zeta(\mathbf{r}) = r^m P_l(x, y, z) e^{-\zeta|\mathbf{r}-\mathbf{R}_A|} \quad (4.76)$$

which is centred around a nucleus located at $\mathbf{R}_A$. Functions of this form are called Slater type orbitals (STO). The parameter ζ , defining the range of the orbital, remains to be determined; P_l is taken the same as for the hydrogen atom. For *atomic* Hartree–Fock calculations, this basis yields accurate results with a restricted basis set size. However, in molecular calculations, integrals involving products of two and four basis functions centred at different nuclei are needed, and these are hard to calculate since the product of the two exponentials,

$$e^{-\zeta_1|\mathbf{r}-\mathbf{R}_A|} e^{-\zeta_2|\mathbf{r}-\mathbf{R}_B|}, \quad (4.77)$$

is a complicated expression in $\mathbf{r}$. A solution might be to evaluate these integrals numerically, but for a large basis set this is impractical.

Another basis set which avoids this problem, was proposed in 1950 by Boys [10], who replaced the simple exponential in (4.76) by a Gaussian function:

$$\chi_\alpha(\mathbf{r}) = P_M(x, y, z) e^{-\alpha(\mathbf{r}-\mathbf{R}_A)^2}. \quad (4.78)$$

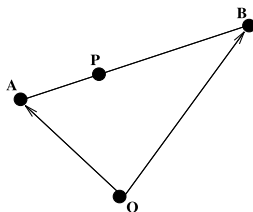


Figure 4.2. Positions in the Gaussian product theorem (4.79).

These functions are called *primitive basis functions* for reasons which will be explained below. These Gaussian type orbitals (GTO) have the nice property that the product of two such functions centred on different nuclei again has a Gaussian form as in (4.78):

$$P_M(x, y, z)e^{-\alpha(\mathbf{r}-\mathbf{R}_A)^2} Q_N(x, y, z)e^{-\beta(\mathbf{r}-\mathbf{R}_B)^2} = R_{N+M}(x, y, z)e^{-(\alpha+\beta)(\mathbf{r}-\mathbf{R}_P)^2}. \quad (4.79)$$

Here, $\mathbf{R}_P$ is the ‘centre of mass’ of the two points $\mathbf{R}_A$ and $\mathbf{R}_B$ with masses α and β :

$$\mathbf{R}_P = \frac{\alpha\mathbf{R}_A + \beta\mathbf{R}_B}{\alpha + \beta} \quad (4.80)$$

(see also Figure 4.2), and R_{M+N} is a polynomial of degree $M+N$. Equation (4.79) is easy to prove; it is known as the ‘Gaussian product theorem’. This theorem makes it possible for the integrals involved in the Hartree–Fock equations to be either calculated analytically or reduced to an expression suitable for fast evaluation on a computer. In Section 4.8 we shall derive some of these integrals.

The polynomials P_M in Eq. (4.78) contain the angular-dependent part of the orbitals, which is given by the spherical harmonics $Y_m^l(\theta, \phi)$. For $l = 0$, these functions are spherically symmetric (no angular dependence) – hence a 1s-orbital (having no nodes) is given as

$$\chi_\alpha^{(s)}(\mathbf{r}) = e^{-\alpha(\mathbf{r}-\mathbf{R}_A)^2}. \quad (4.81)$$

Note that we need not normalise our basis functions: the overlap matrix will ensure proper normalisation of the final molecular orbitals. For $l = 1$, the L_z -quantum number m can take on the three different values 1, 0 and -1 , so there are three p-orbitals, and an explicit GTO representation is

$$\chi_\alpha^{(p_x)}(\mathbf{r}) = xe^{-\alpha(\mathbf{r}-\mathbf{R}_A)^2}, \quad (4.82)$$

and similarly for p_y and p_z . Proceeding in the same fashion for $l = 2$, we find six quadratic factors x^2, y^2, z^2, xy, yz and xz before the Gaussian exponential, but there

are only five d-states! This paradox is solved by noticing that the linear combination

$$(x^2 + y^2 + z^2)e^{-\alpha(\mathbf{r}-\mathbf{R}_A)^2} \quad (4.83)$$

has the symmetry of an s-orbital, and therefore, instead of x^2 , y^2 and z^2 , only the orbitals $x^2 - y^2$ and $3z^2 - r^2$ are used (any independent combination is allowed), thus arriving at five d-states.

GTOs are widely used for molecular calculations, and from now on we shall restrict ourselves to basis functions of this form. The simplest basis consists of one GTO per atomic orbital; it is called a *minimal basis*. The parameter α in the exponent must somehow be chosen such that the GTO fits the atomic orbital in an optimal way. However, since there is only one parameter to be fitted in the minimal basis, this will give poor results, and in general more GTOs per atomic orbital are used. This means that a 1s basis orbital is now given as a linear combination of Gaussian functions:

$$\sum_p D_p e^{-\alpha_p r^2}. \quad (4.84)$$

As the parameters α_p occur in the exponent, determination of the best combination (D_p, α_p) is a nonlinear optimisation problem. We shall not go into details of solving such a problem (see Ref. [11]), but discuss the criterion according to which the best values for (D_p, α_p) are selected. A first approach is to take Hartree–Fock orbitals resulting from an *atomic* calculation, perhaps determined using Slater type orbitals, and to fit the form (4.84) to these orbitals. A second way is to perform the *atomic* Hartree–Fock calculation using Gaussian primitive basis functions and determine the optimal set as a solution to the nonlinear variational problem in the space given by (4.84).

Suppose we have determined the optimal set (D_p, α_p), then there are in principle two options for constructing the basis set. The first option is to incorporate for each exponential parameter α_p , the Gaussian function

$$e^{-\alpha_p(\mathbf{r}-\mathbf{R}_A)^2} \quad (4.85)$$

into the basis, that is, the values of the D_p -parameters are relaxed since the prefactors of the Gaussian primitive functions can vary at will with this basis. A second approach is to take the linear combination (4.84) with the optimal set of (D_p, α_p) as a *single* basis function and add it to the basis set, i.e. keeping the D_p fixed as well as the α_p . If we have optimised the solution (4.84) using four primitive functions, the second option yields a basis four times smaller than the first but, because of its lack of flexibility (remember the D_p are kept fixed), it will result in lower accuracy. The procedure of taking fixed linear combinations of Gaussian primitive functions as a single basis function is called *contraction*; the basis set is called a *contracted set*. The difference between the GTOs from which the basis functions are constructed

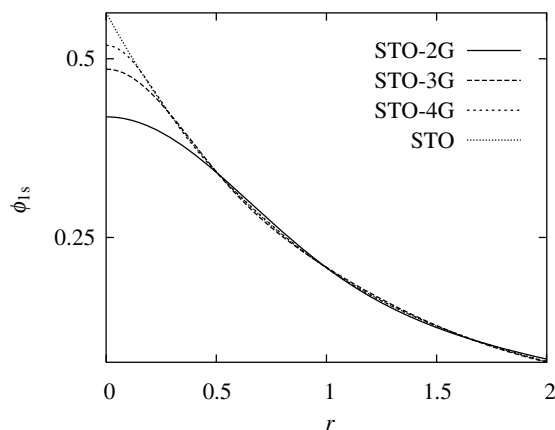


Figure 4.3. Approximation of a 1s Slater orbital (STO) by STO-2G up to STO-4G basis functions.

and the basis functions themselves is the reason why the GTOs were called *primitive* basis functions above: they are used to build contracted basis functions. If all N primitive basis functions are contracted into one normal basis function, the basis set is called a STO- NG basis, denoting that N GTOs have been used to fit a Slater type orbital. Figure 4.3 shows how a Slater orbital is approximated by STO- NG basis functions for $N = 2, 3, 4$. Note that deviations from the exact Slater orbital are only noticeable for small r , and these are usually suppressed, as in volume integrals the integrands are weighted with a factor r^2 .

In most basis sets, an intermediate approach is taken, in which the set of primitive Gaussians is split into two (or more) parts. The primitive Gaussian functions in each such part are contracted into one basis function. A common example is the STO-31 basis set, in which the three basis functions with shortest range (i.e. highest α_p) are contracted (using the corresponding D_p s) into one basis function and the remaining one (with the longest range) is taken as a second basis function. The reason for splitting the basis in this way is that the perturbation of the orbitals by the environment will affect primarily the long-range tails, and leave the short-range part essentially unchanged.

There exist many other basis sets, like split-basis sets, in which from every primitive Gaussian two new ones are constructed, one with a range slightly shorter than in the original function (α slightly larger) and another with a slightly longer range (α slightly smaller), so that it is possible for an orbital in a molecule to assume a slightly contracted or expanded form with respect to the atomic orbital. For more details concerning the various basis sets, see Ref. [12].

An important consideration is the symmetry of the orbitals which should be taken into account. As an example, consider a HF calculation for H_2 . The starting

point for this calculation is the atomic orbitals of the ground state of the isolated H-atom. In the latter, only the (spherically symmetric) 1s orbitals are filled, and in the H₂ molecule, these orbitals will merge into a single molecular orbital which is given by the sum of the two atomic orbitals plus a correction containing a substantial contribution from the atomic p_z-orbitals (the axis connecting the two nuclei is taken to be the z-axis). This shows that it is sensible to include these p_z-orbitals in the basis, even though they are not occupied in the ground state of the isolated atom. Such basis states, which are included in the basis set to make it possible for the basis to represent the polarisation of the atom by its anisotropic environment, are called *polarisation orbitals*. When calculating (dipole, quadrupole, ...) moments by switching on an electric or magnetic field and studying the response of the orbitals to this field, it is essential to include such states into the basis.

4.7 The structure of a Hartree–Fock computer program

In this section we describe the structure of a typical computer program for solving the Roothaan equations. The program for the helium atom as described in [Section 4.3.2](#) contained most of the features present in Hartree–Fock programs already – the treatment given here is a generalisation for arbitrary molecules.

As the most time-consuming steps in this program involve the two-electron integrals, we consider these in some detail in the next subsection. The general scheme of the HF program is then given in [Section 4.7.2](#).

4.7.1 The two-electron integrals

The two-electron integrals are the quantities

$$\langle pr | g | qs \rangle = \int d^3r d^3r' \chi_p(\mathbf{r}) \chi_r(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|} \chi_q(\mathbf{r}) \chi_s(\mathbf{r}'). \quad (4.86)$$

(Do not confuse the *label* r with the orbital coordinate $\mathbf{r}$!) The two-electron matrix elements obey the following symmetry relations:

$$p \leftrightarrow q \quad r \leftrightarrow s \quad p, q \leftrightarrow r, s. \quad (4.87)$$

This implies that, starting with K basis functions, there are roughly $K^4/8$ two-electron matrix elements, since each of the symmetries in (4.87) allows the reduction of the number of different matrix elements to be stored by a factor of about 2.

The subset of two-electron matrix elements to be calculated can be selected in the following way. Because p and q can be interchanged, we can take $p \geq q$. As the pair p, q can be interchanged with the pair r, s , we may also take $p \geq r$. The range of s depends on whether $p = r$ or not. If $p \neq r$, we have $s \leq r$, but for $p = r$, the

fact that p, q can be interchanged with r, s means that the range of s can be restricted to $s \leq q$. All in all, a loop over the different two-electron matrix elements, using K basis vectors and taking into account all symmetries, reads:

```

FOR  $p = 1$  TO  $K$  DO
  FOR  $q = 1$  TO  $p$  DO
    FOR  $r = 1$  TO  $p - 1$  DO
      FOR  $s = 1$  TO  $r$  DO
        .....
        .....
      END FOR
    END FOR
     $r = p$ 
    FOR  $s = 1$  TO  $q$  DO
      .....
      .....
    END FOR
  END FOR
END FOR

```

Suppose we are dealing with a molecule consisting of 6 atoms, each having 12 basis functions, then $K^4/8$ is about 3.4 million. It is not always possible to keep all these numbers in core-memory, in which case they must be stored on disk. In earlier days (before about 1975) this was done by storing them in a prescribed order in a one-dimensional array whose index could be converted into the numbers p, q, r and s . This was inefficient, however, since because of symmetry, many matrix elements may vanish or be equal, or matrix elements may vanish because they involve basis orbitals lying far apart, i.e. having negligible overlap, so nowadays only the non-negligible matrix elements are stored together with the values of the corresponding indices p, q, r and s .

4.7.2 General scheme of the HF program

First, we give an outline and then fill in the details [6, 13].

- Input data;
- Determine matrices;
- Bring overlap matrix to unit form;
- Make a first guess for the density matrix;
- REPEAT
 - Calculate Coulomb and exchange contributions to the Fock matrix;
 - Add these terms to the uncoupled one-electron Hamiltonian h_{pq} ;

- Diagonalise the Fock matrix;
- Construct a new density matrix $\mathbf{P}$ from the eigenvectors found;
- UNTIL converged;
- Output.

Input data: The user provides the coordinates of the positions of the nuclei $\mathbf{R}_n$, the atomic numbers Z_n , the total number of electrons N and a basis set to be used (most programs have several basis sets built into them).

Determine matrices: All matrices that do not depend on the eigenvectors $\mathbf{C}_k$ can be determined here: the overlap matrix S_{pq} , the uncoupled one-electron Hamiltonian h_{pq} , and the two-electron integrals $\langle pr|g|qs \rangle$.

Bring overlap matrix to unit form: This is the procedure described in Section 3.3. It results in a matrix $\mathbf{V}$ defining the basis transformation which brings $\mathbf{S}$ to unit form.

Make a first guess for the density matrix: It is possible to take $P_{pq} = 0$ as a first guess: this implies that the electrons feel the nuclei but not each other. It is then expected that the HF self-consistency procedure will converge with this choice. More elaborate guesses for P can be used in order to increase the probability of convergence, but we shall not go into this here.

Calculate Coulomb and exchange contributions to the Fock matrix: *This is the most time-consuming step in the program!*

The exchange and Coulomb contributions are stored in a matrix $\mathbf{G}$ according to the following formula (see also (4.70)):

$$G_{pq} = \sum_{rs} P_{rs} \left[\langle pr|g|qs \rangle - \frac{1}{2} \langle pr|g|sq \rangle \right]. \quad (4.88)$$

This is done in a loop of the type displayed in the previous subsection. For each combination of the indices p, q, r and s occuring in this loop it must be checked to which elements of $\mathbf{G}$ the corresponding two-electron matrix element contributes. This, however, depends on which of the indices coincide. When all four indices are different for example, one obtains the following contributions to the matrix $\mathbf{G}$:

$$\begin{aligned} G_{pq} &= G_{pq} + 2\langle pr|g|qs \rangle P_{rs} \\ G_{rs} &= G_{rs} + 2\langle pr|g|qs \rangle P_{pq} \\ G_{ps} &= G_{ps} - 1/2\langle pr|g|qs \rangle P_{qr} \\ G_{qr} &= G_{qr} - 1/2\langle pr|g|qs \rangle P_{ps} \\ G_{pr} &= G_{pr} - 1/2\langle pr|g|qs \rangle P_{qs} \\ G_{qs} &= G_{qs} - 1/2\langle pr|g|qs \rangle P_{pr}. \end{aligned} \quad (4.89)$$

Counting all different configurations of the indices, like $p = r \neq q, s$ and $q \neq s$, and so on, 14 different cases are found, and for each of these, the analogue of (4.89) has to be worked out. We shall not do this here but leave it as an exercise to the reader.

Add $\mathbf{G}$ and $\mathbf{h}$: The Fock matrix is simply the sum of the matrix $\mathbf{G}$ which was discussed above, and the uncoupled one-electron matrix $\mathbf{h}$:

$$F_{pq} = h_{pq} + G_{pq}. \quad (4.90)$$

Diagonalise the Fock matrix: To diagonalise $\mathbf{F}$, we transform the Roothaan equation (4.64) with the help of the matrix $\mathbf{V}$ to an ordinary eigenvalue as in Section 3.3:

$$\mathbf{F}' = \mathbf{V}^\dagger \mathbf{F} \mathbf{V}, \quad (4.91)$$

and then we must solve

$$\mathbf{F}' \mathbf{C}' = \epsilon \mathbf{C}' \quad (4.92)$$

and transform the eigenvectors $\mathbf{C}'$ back to the original ones:

$$\mathbf{C} = \mathbf{V} \mathbf{C}'. \quad (4.93)$$

For diagonalising the Fock matrix, we may use the Givens–Householder QR method discussed in Section 7.2.1 of Appendix A, which can be found in the Lapack/Atlas library.

Construct a new density matrix $\mathbf{P}$: From the eigenvectors $\mathbf{C}$ found in diagonalising the Fock matrix, a new density matrix can be constructed which is the ingredient of the Fock matrix in the next iteration. It is important to realise that the vectors $\mathbf{C}_k$ used in constructing the density matrix should be normalised according to

$$1 = \langle \psi_k | \psi_k \rangle = \sum_{pq} C_{pk} \langle p | q \rangle C_{qk} = \mathbf{C}_k \mathbf{S} \mathbf{C}_k. \quad (4.94)$$

Converged: The criterion for convergence of the iterative process is the amount by which the Fock levels and/or the basis functions change from one iteration to the next. A typical criterion is that the Fock levels have converged to within a small margin, such as 10^{-6} a.u. Sometimes, however, the process does not converge but ends up oscillating between two or more configurations, or it diverges. This is often because that the initial guess for P was too far off the converged value. It might then help to use *mixing*: if the changes in the density matrix from one iteration to another are too large, one takes for the density matrix in the next iteration a weighted average of the last and the previous density matrix:

$$P_{pq}^{\text{new}} = \alpha P_{pq}^{\text{last}} + (1 - \alpha) P_{pq}^{\text{previous}}, \quad 0 < \alpha < 1. \quad (4.95)$$

The convergence can finally be enhanced by extrapolating the values of the density matrix. Various extrapolation schemes are used, one of the most popular being Pulay's DIIS scheme (see [Section 9.4](#)) [14].

Output: Output of the program are the Fock levels and corresponding eigenstates. From these data, the total energy can be determined:

$$E = \frac{1}{2} \left[\sum_{rs} h_{rs} P_{rs} + \sum_k \epsilon_k \right] + E_{\text{nucl}}. \quad (4.96)$$

ϵ_k are the Fock levels and E_{nucl} represents the electrostatic nuclear repulsion energy which is determined by the nuclear charges Z_n and positions R_n . Invoking Koopman's theorem (see [Section 4.5.3](#)), the Fock levels may be interpreted as electron removal or addition energies.

In Problem 4.12, the hydrogen molecule is treated again in Hartree–Fock rather than in Hartree theory, as is done in Problem 4.9.

*4.8 Integrals involving Gaussian functions

In this section, we describe some simple calculations of integrals involving two or four GTOs. We restrict ourselves to 1s-functions; for matrix elements involving higher l -values, see [Refs. \[15, 16\]](#). As noticed already in [Section 4.6.2](#), the central result which is used in these calculations is the Gaussian product theorem: denoting the Gaussian function $\exp(-\alpha|\mathbf{r} - \mathbf{R}_A|^2)$ by $g_{1s,\alpha}(\mathbf{r} - \mathbf{R}_A)$, we have:

$$g_{1s,\alpha}(\mathbf{r} - \mathbf{R}_A) g_{1s,\beta}(\mathbf{r} - \mathbf{R}_B) = K g_{1s,\gamma}(\mathbf{r} - \mathbf{R}_P), \quad (4.97)$$

with

$$\begin{aligned} K &= \exp \left[-\frac{\alpha\beta}{\alpha + \beta} |\mathbf{R}_A - \mathbf{R}_B|^2 \right] \\ \gamma &= \alpha + \beta \\ \mathbf{R}_P &= \frac{\alpha\mathbf{R}_A + \beta\mathbf{R}_B}{\alpha + \beta}. \end{aligned} \quad (4.98)$$

From now on, we shall use the Dirac notation:

$$g_{1s,\alpha}(\mathbf{r} - \mathbf{R}_A) = |1s, \alpha, A\rangle. \quad (4.99)$$

The overlap integral: The overlap matrix for two 1s-functions can be calculated directly using (4.97):

$$\begin{aligned} \langle 1s, \alpha, A | 1s, \beta, B \rangle &= 4\pi \int d\mathbf{r} r^2 K e^{-\gamma r^2} \\ &= \left(\frac{\pi}{\alpha + \beta} \right)^{3/2} \exp \left[-\frac{\alpha\beta}{\alpha + \beta} |\mathbf{R}_A - \mathbf{R}_B|^2 \right]. \end{aligned} \quad (4.100)$$

The kinetic integral: This is given by

$$\langle 1s, \alpha, A | -\nabla^2 | 1s, \beta, B \rangle = \int d^3r \nabla e^{-\alpha(\mathbf{r}-\mathbf{R}_A)^2} \nabla e^{-\beta(\mathbf{r}-\mathbf{R}_B)^2} \quad (4.101)$$

where we have used Green's theorem. Working out the gradients and using the Gaussian product theorem, we arrive at

$$\langle 1s, \alpha, A | -\nabla^2 | 1s, \beta, B \rangle = 4\alpha\beta \int d^3r (\mathbf{r} - \mathbf{R}_A)(\mathbf{r} - \mathbf{R}_B) K e^{-\gamma(\mathbf{r}-\mathbf{R}_P)^2} \quad (4.102)$$

with γ , K and $\mathbf{R}_P$ given above. Substituting $\mathbf{u} = \mathbf{r} - \mathbf{R}_P$ and using the fact that integrals antisymmetric in $\mathbf{u}$ vanish, we arrive at:

$$\begin{aligned} & \langle 1s, \alpha, A | -\nabla^2 | 1s, \beta, B \rangle \\ &= 16\alpha\beta K \pi \left[\int_0^\infty du u^4 e^{-\gamma u} + (\mathbf{R}_P - \mathbf{R}_A) \cdot (\mathbf{R}_P - \mathbf{R}_B) \int_0^\infty du u^2 e^{-\gamma u} \right] \\ &= \frac{\alpha\beta}{\alpha + \beta} \left[6 - 4 \frac{\alpha\beta}{\alpha + \beta} |\mathbf{R}_A - \mathbf{R}_B|^2 \right] \\ &\quad \times \left(\frac{\pi}{\alpha + \beta} \right)^{3/2} \exp[-\alpha\beta/(\alpha + \beta) |\mathbf{R}_A - \mathbf{R}_B|^2]. \end{aligned} \quad (4.103)$$

The nuclear attraction integral: This integral is more difficult than the previous ones, because after applying the Gaussian product theorem we are still left with the $1/r$ Coulomb term of the nucleus (whose position does in general not coincide with the centre of the orbital). To reduce the integral to a simpler form, we use Fourier transforms:

$$\hat{f}(\mathbf{k}) = \int d^3r f(\mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{r}}. \quad (4.104)$$

The inverse transformation is given by

$$f(\mathbf{r}) = (2\pi)^{-3} \int d^3k \hat{f}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}}. \quad (4.105)$$

The Dirac delta-function can be written as

$$\delta(\mathbf{r}) = (2\pi)^{-3} \int d^3k e^{i\mathbf{k} \cdot \mathbf{r}}. \quad (4.106)$$

The Coulomb integral is given by

$$\langle 1s, \alpha, A | -Z/r_C | 1s, \beta, B \rangle = -Z \int d^3r K e^{-\gamma|\mathbf{r}-\mathbf{R}_P|^2} |\mathbf{r} - \mathbf{R}_C|^{-1}. \quad (4.107)$$

The Fourier transform of $1/r$ is $4\pi/k^2$, as can be seen for example by Fourier transforming the Poisson equation

$$-\nabla^2 \frac{1}{r} = 4\pi \delta(r). \quad (4.108)$$

Furthermore, the Fourier transform of $\exp(-\gamma r^2)$ is $(\pi/\gamma)^{3/2} \exp(-k^2/4\gamma)$, so substituting these transforms into (4.107), we obtain

$$\begin{aligned} \langle 1s, \alpha, A | -Z/r_C | 1s, \beta, B \rangle \\ = -Z(2\pi)^{-6} \left(\frac{\pi}{\gamma} \right)^{3/2} \int d^3r d^3k_1 d^3k_2 K e^{-k_1^2/(4\gamma)} e^{i\mathbf{k}_1 \cdot (\mathbf{r} - \mathbf{R}_P)} \\ \times 4\pi k_2^{-2} e^{i\mathbf{k}_2 \cdot (\mathbf{r} - \mathbf{R}_C)}. \end{aligned} \quad (4.109)$$

In this equation, the expression (4.106) for the delta-function for $\mathbf{k}_1 + \mathbf{k}_2$ is recognised, and this transforms the integral into

$$\begin{aligned} \langle 1s, \alpha, A | -Z/r_C | 1s, \beta, B \rangle \\ = -ZK(2\pi^2)^{-1} \left(\frac{\pi}{\gamma} \right)^{3/2} \int d^3k e^{-k^2/(4\gamma)} k^{-2} e^{-i\mathbf{k} \cdot (\mathbf{R}_P - \mathbf{R}_C)}. \end{aligned} \quad (4.110)$$

Integrating over the angular variables leads to

$$\begin{aligned} \langle 1s, \alpha, A | -Z/r_C | 1s, \beta, B \rangle = \mathcal{N} \int_0^\infty dk e^{-k^2/(4\gamma)} 1/k \sin(k|\mathbf{R}_P - \mathbf{R}_C|); \\ \mathcal{N} = -2ZK(\pi|\mathbf{R}_P - \mathbf{R}_C|)^{-1}(\pi/\gamma)^{3/2}. \end{aligned} \quad (4.111)$$

The integral (without $\mathcal{N}$) can be rewritten as

$$I(x) \equiv \frac{1}{2} \int_0^x dy \int_{-\infty}^\infty dk e^{-k^2/(4\gamma)} \cos(ky) \quad (4.112)$$

with $x = |\mathbf{R}_P - \mathbf{R}_C|$. The integral over k is easy, and the result is

$$I(x) = \frac{1}{2} \sqrt{\pi/\gamma} \int_0^x dy e^{-\gamma y^2}. \quad (4.113)$$

So, finally, we have

$$\begin{aligned} \langle 1s, \alpha, A | -Z/r_C | 1s, \beta, B \rangle \\ = -2\pi KZ\gamma^{-1}(\gamma^{1/2}|\mathbf{R}_P - \mathbf{R}_C|)^{-1} \int_0^{\gamma^{1/2}|\mathbf{R}_P - \mathbf{R}_C|} dy e^{-y^2}, \end{aligned}$$

and, using the definition

$$F_0(t) = t^{-1/2} \int_0^{t^{1/2}} dy e^{-y^2}, \quad (4.114)$$

the final result can be rewritten as

$$\begin{aligned} \langle 1s, \alpha, A | -Z/r_C | 1s, \beta, B \rangle \\ = -2\pi Z(\alpha + \beta)^{-1} \exp[-\alpha\beta/(\alpha + \beta)|\mathbf{R}_A - \mathbf{R}_B|^2] \\ \times F_0[(\alpha + \beta)|\mathbf{R}_P - \mathbf{R}_C|^2]. \end{aligned} \quad (4.115)$$

The function $F_0(t)$ can be evaluated using the error function erf , which is available in most high-level programming languages as an intrinsic function. The error function is defined by

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x dx' e^{-x'^2}. \quad (4.116)$$

If your compiler does not have the error function as an intrinsic, you can calculate it using a recursive procedure. The function $F_0(t)$ is then considered as one in a series of functions defined as

$$F_m(t) = \int_0^1 \exp(-ts^2) s^{2m} ds. \quad (4.117)$$

The following recursion relation for $F_m(t)$ can easily be derived via partial integration:

$$F_m(x) = \frac{e^{-x} + 2xF_{m+1}(x)}{2m+1}. \quad (4.118)$$

This recursion is stable only if performed downward (see [Appendix A2](#)).

The two-electron integral: This has the form:

$$\begin{aligned} & \langle 1s, \alpha, A; 1s, \beta, B | g | 1s, \gamma, C; 1s, \delta, D \rangle \\ &= \int d^3r_1 d^3r_2 e^{-\alpha|\mathbf{r}_1-\mathbf{R}_A|^2} e^{-\beta|\mathbf{r}_2-\mathbf{R}_B|^2} \frac{1}{r_{12}} e^{-\gamma|\mathbf{r}_1-\mathbf{R}_C|^2} e^{-\delta|\mathbf{r}_2-\mathbf{R}_D|^2}. \end{aligned} \quad (4.119)$$

Using the Gaussian product theorem, we can write the Gaussian functions depending on $\mathbf{r}_1$ and $\mathbf{r}_2$ as new 1s-functions with exponential parameters ρ and σ and centres $\mathbf{R}_P$ (of $\mathbf{R}_A$ and $\mathbf{R}_C$) and $\mathbf{R}_Q$ (of $\mathbf{R}_B$ and $\mathbf{R}_D$):

$$\begin{aligned} & \langle 1s, \alpha, A; 1s, \beta, B | g | 1s, \gamma, C; 1s, \delta, D \rangle \\ &= \exp[-\alpha\gamma/(\alpha+\gamma)|\mathbf{R}_A-\mathbf{R}_C|^2 - \beta\delta/(\beta+\delta)|\mathbf{R}_B-\mathbf{R}_D|^2] \\ &\quad \times \int d^3r_1 d^3r_2 e^{-\rho|\mathbf{r}_1-\mathbf{R}_P|^2} \frac{1}{r_{12}} e^{-\sigma|\mathbf{r}_2-\mathbf{R}_Q|^2}. \end{aligned} \quad (4.120)$$

Calling $\mathcal{M}$ the factor in front of the integral, and replacing the Gaussian exponentials in the integrand by their Fourier transforms, and similarly for the $1/r_{12}$ term, we obtain

$$\begin{aligned} & \langle 1s, \alpha, A; 1s, \beta, B | g | 1s, \gamma, C; 1s, \delta, D \rangle \\ &= \mathcal{M} (2\pi)^{-9} \int d^3r_1 d^3r_2 d^3k_1 d^3k_2 d^3k_3 (\pi/\rho)^{3/2} e^{-k_1^2/4\rho} e^{i\mathbf{k}_1 \cdot (\mathbf{r}_1-\mathbf{R}_P)} \\ &\quad \times 4\pi k_2^{-2} e^{i\mathbf{k}_2 \cdot (\mathbf{r}_1-\mathbf{r}_2)} (\pi/\sigma)^{3/2} e^{-k_3^2/4\sigma} e^{i\mathbf{k}_3 \cdot (\mathbf{r}_2-\mathbf{R}_Q)}. \end{aligned} \quad (4.121)$$

Table 4.1. *Bond lengths in atomic units for three different molecules. Hartree–Fock (HF) and experimental results are shown.*

Molecule	HF	Expt.
H ₂	1.385	1.401
N ₂	2.013	2.074
CO	1.914	1.943

Data taken from [Ref. \[6\]](#).

The integrals over $\mathbf{r}_1$ and $\mathbf{r}_2$ yield two delta-functions in $\mathbf{k}_1$ and $\mathbf{k}_2$, and [Eq. \(4.121\)](#) transforms into

$$\begin{aligned} &\langle 1s, \alpha, A; 1s, \beta, B | g | 1s, \gamma, C; 1s, \delta, D \rangle \\ &= 4\pi \mathcal{M} (2\pi)^{-3} (\pi^2 / \rho\sigma)^{3/2} \int d^3k \, k^{-2} e^{-k^2(\rho+\sigma)/(4\rho\sigma)} e^{i\mathbf{k}(\mathbf{R}_P - \mathbf{R}_Q)}. \end{aligned} \quad (4.122)$$

We have already encountered this integral in the previous subsection. The final result is now

$$\begin{aligned} &\langle 1s, \alpha, A; 1s, \beta, B | g | 1s, \gamma, C; 1s, \delta, D \rangle \\ &= \frac{2\pi^{(5/2)}}{(\alpha + \gamma)(\beta + \delta)(\alpha + \beta + \gamma + \delta)^{1/2}} \\ &\quad \times \exp[-\alpha\gamma/(\alpha + \gamma)|\mathbf{R}_A - \mathbf{R}_C|^2 - \beta\delta/(\beta + \delta)|\mathbf{R}_B - \mathbf{R}_D|^2] \\ &\quad \times F_0 \left[\frac{(\alpha + \gamma)(\beta + \delta)}{(\alpha + \beta + \gamma + \delta)} |\mathbf{R}_P - \mathbf{R}_Q|^2 \right]. \end{aligned} \quad (4.123)$$

4.9 Applications and results

After having considered the implementation of the Hartree–Fock in a computer program, we now present some results of HF calculations for simple molecules [\[6, 17\]](#). As the HF calculations yield an energy for a static configuration of nuclei, it is possible to find the stable configuration by varying the positions of the nuclei and calculating the corresponding energies – the lowest energy corresponds to the stable configuration. In this way, the equilibrium bond lengths of diatomic molecules can be determined. In [Table 4.1](#), HF results are shown for these bond lengths, together with experimental results. The table shows good agreement between the two. The same holds for bond angles, given in [Table 4.2](#). It is also possible to calculate the

Table 4.2. *Bond angles in degrees for H₂O and NH₃. The angles are those of the H–O–H and H–N–H chains respectively. Hartree–Fock (HF) and experimental results are shown.*

Molecule	HF	Expt.
H ₂ O	107.1	104.5
NH ₃	108.9	106.7

Data taken from [Ref. \[17\]](#).

Table 4.3. *Dissociation energies in atomic units for LiF and NaBr. Hartree–Fock (HF) and experimental results are shown.*

Molecule	HF	Expt
LiF	0.2938	0.2934
NaBr	0.1978	0.2069

Data taken from [Ref. \[17\]](#).

energies needed to dissociate diatomic molecules (see [Table 4.3](#)) and again good agreement is found with experiment.

Koopman’s theorem can be used to calculate ionisation potentials, that is, the minimum energy needed to remove an electron from the molecule. Comparing the results in [Table 4.4](#) for the ionisation potentials calculated via Koopman’s theorem with those of the previous tables, it is seen that the approximations involved in this ‘theorem’ are more severe than those of the Hartree–Fock theory, although agreement with experiment is still reasonable.

***4.10 Improving upon the Hartree–Fock approximation**

The Hartree–Fock approximation sometimes yields unsatisfactory results. This is of course due to Coulomb correlations not taken into account in the Hartree–Fock formalism. There exists a systematic way to improve on Hartree–Fock by constructing a many-electron state as a linear combination of Slater determinants (remember the Slater determinants span the N -electron Hilbert space of many-electron wave

Table 4.4. *Ionisation potentials in atomic units for various molecules. Results obtained via Koopman’s theorem and experimental results are shown.*

Molecule	Koopman	Expt
H ₂	0.595	0.584
CO	0.550	0.510
H ₂ O	0.507	0.463

Data taken from [Ref. \[6\]](#).

functions as mentioned at the end of [Section 4.4](#)). These determinants are constructed from the ground state by excitation: the first determinant is the Hartree–Fock ground state and the second one is the first excited state (within the spectrum determined self-consistently for the ground state) and so on. The matrix elements of the Hamiltonian between these Slater determinants are calculated and the resulting Hamilton matrix (which has a dimension equal to the number of Slater determinants taken into account) is diagonalised. The resulting state is then a linear combination of Slater determinants

$$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N) = \sum_n \alpha_n \Psi_{AS}^{(n)}(\mathbf{x}_1, \dots, \mathbf{x}_N) \quad (4.124)$$

and its energy will be lower than the Hartree–Fock ground state energy. This is a time-consuming procedure so that for systems containing many electrons, only a limited number of determinants can be taken into account. This is the configuration interaction (CI) method. In simple systems, for which this method allows very high accuracy to be achieved, excellent agreement with experimental results can be obtained. The CI method is in principle exact (within the Born–Oppenheimer approximation), but since for a finite basis set the Fock spectrum is finite, only a finite (though large) number of Slater determinants is possible within that basis set. A CI procedure in which all possible Slater determinants possible within a chosen basis set are taken into account is called ‘full CI’. For most systems, full CI is impossible because of the large number of Slater determinants needed, but it is sometimes possible to obtain an estimate for the full CI result by extrapolating results for larger and larger numbers of Slater determinants.

As an illustration, we show bond lengths and correlation energies for H₂ and H₂O in [Tables 4.5](#) and [4.6](#). The correlation energy is defined as the difference between the Hartree–Fock and the exact energy. For small systems such as H₂, the electronic structure can be calculated taking the electron correlation fully into account (but

Table 4.5. *Correlation energies in atomic units for H_2 and H_2O .*

Molecule	CI	Exact
H_2	−0.039 69	−0.0409
H_2O	−0.298	−0.37

Data taken from [Ref. \[6\]](#).

Table 4.6. *Bond lengths in atomic units for H_2 and H_2O .*

Molecule	HF	CI	Exact	Experiment
H_2	1.385	1.396	1.401	
H_2O	1.776	1.800		1.809

Data taken from [Ref. \[6\]](#). Exact results were obtained by variational calculus [\[18\]](#). Experimental results are from [Ref. \[19\]](#).

within the Born–Oppenheimer approximation) by a variational method using basis functions depending on the positions of both electrons [\[18\]](#). The CI results are excellent for both cases.

In CI, the spin-orbitals from which the Slater determinants are built are the eigenstates of the Fock operator as determined self-consistently for the ground state. As only a restricted number of determinants can be taken into account, the dimension of the subspace spanned by the Slater determinants is quite limited. If, within these Slater determinants, the orbitals are allowed to vary by relaxing the ground state coefficients of the basis functions, this subspace can be increased considerably. In the so-called multi-configuration self-consistent field theory (MCSCF), this process is carried out, but because of the large amount of variation possible this leads to a huge numerical problem. Finally, perturbative analysis allows correlation effects to be taken into account [\[5, 6\]](#).

Exercises

- 4.1 In this problem we show that the large masses of the nuclei compared with those of the electrons lead to the Born–Oppenheimer approximation.

The wave function Ψ of a collection of electrons and nuclei depends on the positions $\mathbf{R}_n$ of the nuclei and $\mathbf{r}_i$ of the electrons (we neglect the spin-degrees of

freedom). For this function we make the following *Ansatz*:

$$\Psi(\mathbf{R}_n, \mathbf{r}_i) = \chi(\mathbf{R}_n) \Phi(\mathbf{r}_i)$$

with $\Phi(\mathbf{r}_i)$ an eigenstate with eigenvalue E_{el} of the N -electron ‘Born–Oppenheimer Hamiltonian’ Eq. (4.2), which in atomic units reads:

$$H_{\text{BO}} = \sum_{i=1}^N -\frac{1}{2} \nabla_i^2 + \frac{1}{2} \sum_{i,j=1; i \neq j}^N \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{n=1}^K \sum_{i=1}^N \frac{Z_n}{|\mathbf{r}_i - \mathbf{R}_n|}.$$

It is clear that Φ and E_{el} depend on the nuclear positions $\mathbf{R}_n$, since the Born–Oppenheimer Hamiltonian does.

Show that substitution of this *Ansatz* into the full Hamiltonian, Eq. (4.1), leads to:

$$\begin{aligned} \Phi(\mathbf{r}) \left[\sum_{n=1}^K -\frac{1}{2M_n} \nabla_n^2 + E_{\text{el}} + \sum_{n,n'=1; n \neq n'}^K \frac{Z_n Z_{n'}}{|\mathbf{R}_n - \mathbf{R}_{n'}|} \right] \chi(\mathbf{R}_n) \\ - \chi(\mathbf{R}_n) \sum_{n=1}^K \frac{1}{2M_n} \nabla_n^2 \Phi(\mathbf{r}_i) - \sum_{n=1}^K \frac{1}{M_n} \nabla_n \chi(\mathbf{R}_n) \cdot \nabla_n \Phi(\mathbf{r}_i) = E \chi(\mathbf{R}_n) \Phi(\mathbf{r}_i), \end{aligned}$$

so that neglecting the last two terms on the left hand side of this equation, we arrive at a Schrödinger equation for the nuclei which contains the electronic degrees of freedom via the electronic energy E_{el} only:

$$\left[\sum_{n=1}^K -\frac{1}{2M_n} \nabla_n^2 + E_{\text{el}} + \sum_{n,n'=1; n \neq n'}^K \frac{Z_n Z_{n'}}{|\mathbf{R}_n - \mathbf{R}_{n'}|} \right] \chi(\mathbf{R}_n) = E_{\text{nuc}} \chi(\mathbf{R}_n).$$

The fact that the term $(1/2M_n) \nabla_n^2 \Phi(\mathbf{r}_i)$ can be neglected can be understood by realising that it is $1/M_n$ times the variation of the kinetic energy of the electrons with the positions of the nuclei. Of course, the core electrons have large kinetic energy, but they feel almost exclusively their own nucleus, hence their kinetic energy is insensitive to variations in the relative nuclear positions. The valence electrons have smaller kinetic energies, so the variation of this energy with nuclear positions will be small too. In a solid, deleting the term $(1/M_n) \nabla_n \chi(\mathbf{R}_n) \cdot \nabla_n \Phi(\mathbf{r}_i)$ means that electron–phonon couplings are neglected, so that some physical phenomena cannot be treated in calculations involving the Born–Oppenheimer approach, although these effects can often be studied perturbatively.

4.2 For a two-electron system, the wave function can be written as

$$\Psi(\mathbf{x}_1, \mathbf{x}_2) = \Phi(\mathbf{r}_1, \mathbf{r}_2) \cdot \chi(s_1, s_2).$$

Because the wave function Ψ is antisymmetric under particle exchange, we may take Φ symmetric in 1 and 2 and χ antisymmetric, or vice versa.

We construct the functions Φ and χ from the orthonormal spatial orbitals $\phi_1(\mathbf{r})$, $\phi_2(\mathbf{r})$ and the spin-up and -down functions $\alpha(s)$ and $\beta(s)$ respectively.

- Write down the antisymmetric wave functions that can be constructed in this way (there are six of them).
- Write down all possible Slater determinants that can be built from the one-electron spin-orbitals consisting of a product of one of the orbitals ϕ_1 and ϕ_2 and a spin-up or -down spinor (you will find six of these determinants too).
- Express the wave functions of part (a) of this problem in those of (b).

4.3 Consider a Slater determinant

$$\begin{aligned}\Psi_{\text{AS}}(\mathbf{x}_1, \dots, \mathbf{x}_N) &= \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_1(\mathbf{x}_1) & \psi_2(\mathbf{x}_1) & \cdots & \psi_N(\mathbf{x}_1) \\ \psi_1(\mathbf{x}_2) & \psi_2(\mathbf{x}_2) & \cdots & \psi_N(\mathbf{x}_2) \\ \vdots & \vdots & \cdots & \vdots \\ \psi_1(\mathbf{x}_N) & \psi_2(\mathbf{x}_N) & \cdots & \psi_N(\mathbf{x}_N) \end{vmatrix} \\ &= \frac{1}{\sqrt{N!}} \sum_{\mathbf{P}} \epsilon_{\mathbf{P}} \mathbf{P} \psi_1(\mathbf{x}_1) \dots \psi_N(\mathbf{x}_N).\end{aligned}$$

The spin-orbitals $\psi_k(\mathbf{x})$ are orthonormal.

- Show that the Slater determinant is normalised, by considering the inner product of two arbitrary terms occurring in the sum of the Slater determinant and then summing over all possible pairs of such terms.
- Show in the same way that the density of electrons with coordinates $\mathbf{x}$, given by:

$$n(\mathbf{x}) = N \int d\mathbf{x}_2 \dots d\mathbf{x}_N |\Psi_{\text{AS}}(\mathbf{x}, \mathbf{x}_2, \dots, \mathbf{x}_N)|^2,$$

can be written in terms of the ψ_k as:

$$n(\mathbf{x}) = \sum_k |\psi_k(\mathbf{x})|^2.$$

Suppose all spin-orbitals can be written as the product of a normalised orbital and a normalised one-particle spinor, what is then the spatial charge density of the electrons (i.e. regardless of the spin)?

- Derive Eqs. (4.35) and (4.36) using the methods employed in (a) and (b).

4.4 Consider the helium atom with two electrons having the *same* spin, represented by the spinor $\alpha(s)$.

- Give the form of the two-electron wave function, expressed in orthonormal spatial orbitals ϕ_1 and ϕ_2 .
- Write down the Schrödinger equation for this system.
- Give an expression for the expectation value of the energy in the orbitals ϕ_i .

4.5 The Hartree–Fock analysis can be performed not only for fermions, but also for bosons. Consider a system consisting of N spin-0 particles in one dimension having spin-orbital coordinates x_i (for spin-0 particles, only the orbital coordinate matters).

The bosons interact via a δ -function potential:

$$H = - \sum_{i=1}^N \frac{\partial^2}{\partial x_i^2} + g \sum_{i>j}^N \delta(x_i - x_j).$$

This means that the particles feel each other only if they are at the same position: they experience an infinitely large attraction with an infinitely short range. Although this problem can be solved exactly, we consider the Hartree–Fock approximation here.

A boson wave function Ψ is symmetric under particle exchange. This means that *all* the particles are in the same orbital ϕ .

(a) Show that the kinetic term of the Hamiltonian has the form:

$$\left\langle \Psi \left| - \sum_{i=1}^N \frac{\partial^2}{\partial x_i^2} \right| \Psi \right\rangle = -N \left\langle \phi \left| \frac{d^2}{dx^2} \right| \phi \right\rangle.$$

(b) Show that the expectation value of the interaction potential is given by

$$\left\langle \Psi \left| g \sum_{i>j} \delta(x_i - x_j) \right| \Psi \right\rangle = \frac{1}{2} N(N-1) \int dx |\phi(x)|^4.$$

(c) Show, by minimisation of the energy-functional

$$\langle E \rangle = \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle}$$

with respect to ϕ , that the Hartree–Fock equation reads

$$\left[-\frac{\partial^2}{\partial x^2} - g(N-1)|\phi(x)|^2 \right] \phi(x) = \epsilon \phi(x).$$

(d) The solution to this last equation is found as

$$\phi(x) = \frac{[(1/8)g(N-1)]^{1/2}}{\cosh[(1/4)g(N-1)x]};$$

$$\epsilon = \frac{1}{16}g^2(N-1)^2.$$

Show that this function indeed satisfies the Hartree–Fock equation.

- 4.6 Consider the Fock spectrum of a many-electron system. In the ground state, the N electrons fill the lowest N levels of this spectrum. Consider the same spectrum, but now with one electron removed from it. This means that the system has been ionised. Show from the expressions for the Fock operator and the energy in terms of the spin-orbitals, Eqs. (4.41) and (4.48), that the ionisation energy is equal to the difference between the sum over the occupied Fock levels ϵ_k in the ground state and the same sum for the ionised state.

It is then clear that the same holds for adding an electron to the ground state, and therefore for moving an electron from level a to level b (by first removing the electron from level a and then adding one in level b).

- 4.7 Consider a Slater determinant constructed from spin-orbitals ψ_k , $k = 1, \dots, N$. A unitary transformation transforms the spin-orbitals ψ_k into new ones, which we denote by ψ'_k :

$$\psi'_k = \sum_{l=1}^N U_{kl} \psi_l.$$

U_{kl} are the elements of the unitary matrix $\mathbf{U}$.

- (a) Show that the new basis is orthonormal.
 (b) Show that the Slater determinant constructed from the new spin-orbitals can be written as the determinant of the product of the matrix

$$\frac{1}{\sqrt{N!}} \begin{pmatrix} \psi_1(\mathbf{x}_1) & \psi_2(\mathbf{x}_1) & \cdots & \psi_N(\mathbf{x}_1) \\ \psi_1(\mathbf{x}_2) & \psi_2(\mathbf{x}_2) & \cdots & \psi_N(\mathbf{x}_2) \\ \vdots & \vdots & & \vdots \\ \psi_1(\mathbf{x}_N) & \psi_2(\mathbf{x}_N) & \cdots & \psi_N(\mathbf{x}_N) \end{pmatrix}$$

and the matrix $\mathbf{U}$.

- (c) Show that the Slater determinant built from ψ_k and that built from ψ'_k are equal up to a complex constant of absolute value unity.
- 4.8 [C] In this problem, the program for calculating the electronic structure of the hydrogen atom (see [Section 3.2.2](#)), is extended to the H_2^+ ion. The H_2^+ ion contains only one electron and the problem is therefore essentially the same as that of [Section 3.2.2](#), the difference being a second nucleus at a distance R from the first one. The global structure of the program is therefore the same, the main difference being that the basis now consists of eight functions: four functions centred around each nucleus. Therefore, all matrices now have dimension 8.

It is important to note that for basis functions centred around one of the two nuclei, the Coulomb attraction of the other nucleus is still important, as is immediately clear from the expression of the Coulomb matrix $\mathbf{A}$:

$$A_{pq} = \int d^3r \chi_p(\mathbf{r}) \left(\frac{1}{|\mathbf{r} - \mathbf{R}_A|} + \frac{1}{|\mathbf{r} - \mathbf{R}_B|} \right) \chi_q(\mathbf{r})$$

where $\mathbf{R}_A$ and $\mathbf{R}_B$ denote the positions of the two nuclei. The integrals for calculating the matrix elements of the various operators can be found in [Section 4.8](#). Write a program to determine the ground state of the H_2^+ ion.

For a distance $1a_0$ between the nuclei, the program should yield an energy (without the Coulomb repulsion between the nuclei) equal to $-1.442\,455$ a.u.

- 4.9 [C] In this problem, the program developed in the previous problem is extended along the lines of the helium ground state calculation of [Section 4.3.2](#) in order to calculate the electronic structure of the hydrogen molecule. This means that a second electron is added to the ionic hydrogen system and we must solve the Hartree equation for a finite basis, [Eq. \(4.14\)](#), self-consistently analogous to the helium calculation.

The matrix to be diagonalised is given by

$$F_{pq} = h_{pq} + \sum_{rs} C_r C_s Q_{prqs}$$

with

$$Q_{prqs} = \langle pr | g | qs \rangle = \int d^3 r_1 d^3 r_2 \phi_p(\mathbf{r}_1) \phi_r(\mathbf{r}_2) \frac{1}{r_{12}} \phi_q(\mathbf{r}_1) \phi_s(\mathbf{r}_2).$$

As we are dealing with two electrons only, we do not have to sum over the different orbitals k .

In Problem 4.8 you have already programmed the expressions for the matrix elements h_{pq} , so these pose no problems. The two-electron matrix elements Q_{prqs} are rather complicated; they are given in [Section 4.8](#). The resulting expressions can be written in the form:

$$\langle \chi_p \chi_r | g | \chi_q \chi_s \rangle = 2 \sqrt{\frac{AB}{\pi(A+B)}} S_{pq} S_{sr} F_0(t).$$

Here, t is defined as

$$t = \frac{(\alpha_p + \alpha_q)(\alpha_r + \alpha_s)}{(\alpha_p + \alpha_q + \alpha_r + \alpha_s)} |\mathbf{R}_A - \mathbf{R}_B|^2;$$

$$\mathbf{R}_A = \frac{\alpha_p \mathbf{R}_p + \alpha_q \mathbf{R}_q}{\alpha_p + \alpha_q}$$

$$\mathbf{R}_B = \frac{\alpha_r \mathbf{R}_r + \alpha_s \mathbf{R}_s}{\alpha_r + \alpha_s},$$

and A and B as

$$A = \alpha_p + \alpha_q$$

$$B = \alpha_r + \alpha_s.$$

S_{pq} is the overlap matrix.

You can now use these matrix elements in a program which has the same structure as that of the helium atom.

Check: for a distance 1 a.u. between the atoms, one finds for the ground state vector $\mathbf{C}$:

$$(0.092\,561\,548\,6, 0.165\,180\,118, 0.120\,122\,665, 0.021\,154\,565\,7,$$

$$0.092\,561\,548\,6, 0.165\,180\,118, 0.120\,122\,665, 0.021\,154\,565\,7)$$

and an energy $-1.078\,547\,61$ (nuclear repulsion $+1$ included!).

- 4.10 In a restricted Hartree–Fock (RHF) calculation using a finite basis $\chi_p(\mathbf{r})$, the kinetic and Coulomb integrals must be calculated. We can gain speed by using the symmetry of these matrices, for example

$$\langle \chi_p | \nabla^2 | \chi_q \rangle = \langle \chi_q | \nabla^2 | \chi_p \rangle.$$

We do not assume other symmetries to be present in the system.

- (a) Suppose the basis contains M basis functions, at least how many of these matrix elements must be calculated?
- (b) How many two-electron matrix elements

$$\langle pr|g|qs\rangle$$

must be calculated?

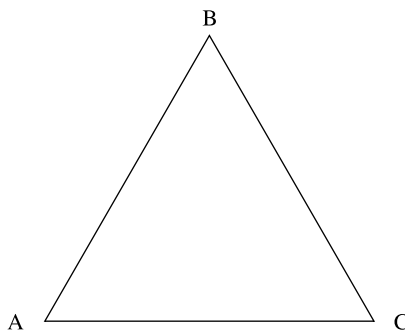
Now suppose that the molecule for which we are performing the calculation consists of three identical atoms, located on an equilateral triangle:

On every atom, we have M basis functions which we denote by $\chi_p^A(\mathbf{r})$ etc., $p = 1, \dots, M$. The basis functions on the different atoms have the same form:

$$\chi_p^A(\mathbf{r}) = \chi_p(\mathbf{r} - \mathbf{R}_A),$$

$$\chi_p^B(\mathbf{r}) = \chi_p(\mathbf{r} - \mathbf{R}_B)$$

and similarly for C .



- (c) How many different kinetic and Coulomb matrix elements must be calculated in this case?
- (d) How many different two-electron matrix elements must be calculated?
- (e) Suppose now that M is very large. What is then the gain in speed when using all symmetries of the matrix elements instead of using no symmetry at all?
- 4.11 Suppose a Hartree–Fock calculation is carried out for a linear chain of K identical atoms and N electrons, where K is a large number. The distance between two successive atoms is a . For each atom, the same set of M basis functions is used. We assume that the overlap between two wave functions centred around two atoms at a distance larger than pa (p is some integer) vanishes.
- (a) How many elements of the Hamilton matrix are nonzero (for large K)?
- (b) How many nonzero two-electron matrix elements $\langle pr|g|qs\rangle$ do we have (for large K)?
- 4.12 [C] In this problem, we modify the hydrogen calculation as carried out in Problem 4.9 to a Hartree–Fock calculation – remember that in the previous version we solved the Hartree equation and not the Hartree–Fock equation. Furthermore, we consider exploiting the symmetry in order to speed up the calculation. As we have

mentioned in [Section 4.3.1](#), the Hartree approach is good enough for the ground state of a two-electron system because the two electrons are described by the *same* orbital: antisymmetry is taken into account via the spins which are opposite. The present program should therefore yield the same result as the previous one but have the structure of the programs dealing with more electrons or excited states.

The main difference from the previous version is that the array Q is replaced by $\tilde{Q}$ in this program, defined in terms of the Q_{pqrs} as follows:

$$\tilde{Q}_{pqrs} = 2Q_{pqrs} - Q_{prsq}.$$

- (a) [C] First of all, you can exploit the symmetry in the calculation of S_{pq} and h_{pq} : both matrices are symmetric, so you only need to calculate the upper or lower triangular elements. Implement this in your program.
- (b) [C] In order to calculate the matrix Q_{pqrs} fast, you can restrict the indices to $p \geq q$, $q \geq r$ and if $p = r$, $q \geq s$, otherwise $r \geq s$, see [Section 4.7](#). For each set of p, q, r, s in these ranges, seven other Q -matrix elements having the same values can then be found because of symmetry. These are: Q_{qrps} , Q_{psqr} , Q_{qspr} , Q_{rpsq} , Q_{sprq} , Q_{rqsp} and Q_{sqrp} .
- (c) [C] Now use the matrix elements $\tilde{Q}_{pqrs}$ instead of Q_{pqrs} and check that you obtain the same results as in the previous program.
- (d) The fact that using $\tilde{Q}$ instead of Q leads to the same results is quite surprising, since it is a different Fock matrix we are diagonalising (you can check this by printing out the Fock matrices of the old and the present program). Show that if the vector $\mathbf{C}$ has converged, the results are equivalent (hint: by inspection of the energy, rather than the Fock matrix).
- (e) Exploit the full symmetry of the two-electron matrix elements by distinguishing all possible cases for p, q, r, s being different or equal. Formulate the equations analogous to [Eq. \(4.89\)](#) for all these cases (there are 14 of them).
- (f) [C] Implement these equations into your program.

References

- [1] B. T. Sutcliffe, 'Fundamentals of computational quantum chemistry,' in *Computational Techniques in Quantum Chemistry and Molecular Physics* (ed. G. H. F. Dierksen, B. T. Sutcliffe, and A. Veillard), Proceedings of the NATO ASI held at Ramsau, Germany, Boston, Reidel, 1975, p. 1.
- [2] D. R. Hartree, *Proc. Camb. Phil. Soc.*, **24** (1928), 89.
- [3] C.-O. Almbladh and A. C. Pedroza, 'Density-functional exchange-correlation potentials and orbital eigenvalues for light atoms,' *Phys. Rev. A*, **29** (1984), 2322–30.
- [4] J. C. Slater, *Quantum Theory of Molecules and Solids*, vol. IV. New York, McGraw-Hill, 1982.
- [5] R. McWeeny, *Methods of Molecular Quantum Mechanics*, 2nd edn. New York, Academic Press, 1989.
- [6] A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry*. London, Macmillan, 1982.
- [7] N. W. Ashcroft and N. D. Mermin, *Solid State Physics*. New York, Holt, Reinhart and Winston, 1976.

- [8] W. J. Hehre, L. Radom, P. R. Schleyer, and J. A. Pople, *Ab Initio Molecular Orbital Theory*. New York, John Wiley, 1986.
- [9] C. A. Coulson and I. Fisher, 'Notes on the molecular orbital treatment of the hydrogen molecule,' *Philos. Mag.*, **40** (1949), 386–93.
- [10] S. F. Boys, 'A general method of calculation for the stationary states of any molecular system,' *Proc. Roy. Soc. (London)*, **200** (1950), 542–54.
- [11] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes*, 2nd edn. Cambridge, Cambridge University Press, 1992.
- [12] R. Poirer, R. Kari, and I. G. Czismadia, *Handbook of Gaussian Basis Sets*. Amsterdam, Elsevier, 1985.
- [13] A. Veillard, 'The logic of SCF procedures,' in *Computational Techniques in Quantum Chemistry and Molecular Physics* (ed. G. H. F. Diercksen, B. T. Sutcliffe, and A. Veillard), Proceedings of the NATO ASI held at Ramsau, Germany, Boston, Reidel, 1975, p. 201.
- [14] P. Pulay, 'Improved SCF convergence acceleration,' *J. Comp. Chem.*, **3** (1982), 556–60.
- [15] E. Clementi and D. R. Davis, 'Electronic structure of large molecular systems,' *J. Comp. Phys.*, **2** (1967), 223–44.
- [16] V. R. Saunders, 'An Introduction to Molecular Integral Evaluation,' in *Computational Techniques in Quantum Chemistry and Molecular Physics* (ed. G. H. F. Diercksen, B. T. Sutcliffe, and A. Veillard), Proceedings of the NATO ASI held at Ramsau, Germany, Boston, Reidel, 1975, pp. 347–92.
- [17] A. Hinchliffe, *Computational Quantum Chemistry*. Chichester, John Wiley & Sons, 1988.
- [18] W. Kolos and L. Wolniewicz, 'Improved theoretical ground-state energy of the hydrogen molecule,' *J. Chem. Phys.*, **49** (1968), 404–10.
- [19] B. J. Rosenberg and I. Shavitt, 'Ab initio SCF and CI studies on the ground state of the water molecule. I. Comparison of CGTO and STO basis sets near the Hartree–Fock limit,' *J. Chem. Phys.*, **63** (1975), 2162–74.